

- 1 Find the complex numbers v and w which satisfy the following simultaneous equations.

$$v^2 - iw + 2 = 0$$

$$v = w + i + 3$$

Give your answers in the form $a + ib$, where a and b are real numbers. [4]

- 2 The path traced by a moving particle is a curve C , given by

$$x = a\theta - a \sin \theta \text{ and } y = a - a \cos \theta,$$

where a is a positive constant and $0 \leq \theta < 2\pi$.

- (a) Show that the gradient of C at a point with parameter θ , is $\cot\left(\frac{1}{2}\theta\right)$. [2]

- (b) Find the equation of the tangent to C at the point where $\theta = \frac{\pi}{3}$ and show that this tangent passes through the point $\left(\frac{1}{3}\pi a, 2a\right)$. [4]

- 3 (a) Given that $y = (\sin^{-1} x)^2$, show that $(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 2$. [2]

- (b) Hence obtain the Maclaurin expansion of y in terms of x , up to and including the term in x^4 . [3]

- (c) By using $x = \frac{1}{2}$, find an approximation for π in surd form. Find the percentage error of this approximation and comment on its accuracy. [3]

- 4 It is given that $f(x) = \sqrt{3} \cos x + \sin x$.

- (a) Write $f(x)$ as $R \cos(x - \alpha)$, where R and α are constants to be found. [2]

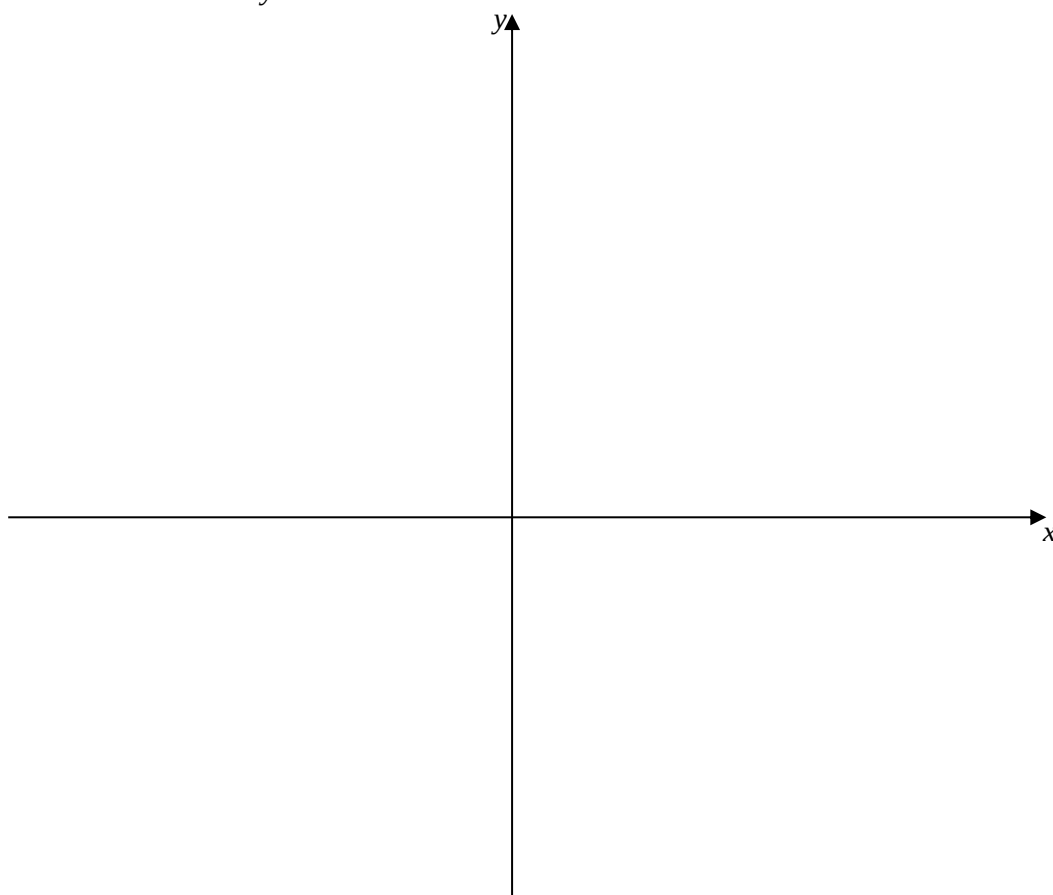
- (b) Find the exact value of $\int_0^{\frac{\pi}{6}} \left(\frac{1}{f(x)}\right)^2 dx$. [2]

- (c) Find the exact value of $\int_0^{\frac{\pi}{12}} \frac{1}{f(2x)} dx$. [3]

- 5 (a) Find $\int \frac{4x-1}{x^2+4x+4} dx$, giving your answer in simplest form. [4]

- (b) Find the value of $\int_0^1 \frac{|4x-1|}{x^2+4x+4} dx$, expressing your answer in the form $p \ln q + r$, where p , q and r are exact constants in simplest form to be found. [3]

- 6 (a) An arithmetic series has first term a and common difference d , where $d \neq 0$. The 1st, 6th and 14th terms of this series are the 1st, 2nd and 3rd terms of a geometric series. Find d in terms of a . [3]
- (b) A geometric series has first term b , where $b > 0$, and common ratio 0.5.
- (i) Find the sum to infinity of this series in terms of b . [1]
- (ii) Find the smallest possible value of n for which the sum of the first n terms of the series differs from the sum of the first $2n$ terms of the series by less than $0.004b$. [4]
- 7 An isosceles triangle ABC , with $AB = AC$, is inscribed in a fixed circle of radius 1 unit and centre O . It is given that angle $BOC = 2\theta$, where θ is acute. Using calculus, show that the area of the triangle ABC is a maximum when it is equilateral. State the maximum value of this area exactly. [8]
- 8 A curve C has equation $y = ax + b + \frac{1}{x-a}$, where a and b are positive real constants such that $a > 1$ and $x \neq a$.
- (a) Sketch C on the axes below stating the equations of any asymptotes and the coordinates of the point where C crosses the y -axis. [4]



- (b) On the same axes, sketch the graph of $(x-a)^2 + (y-a^2-b)^2 = r^2$, where r is a positive constant, such that it intersects C at more than 2 points. $y = \pm\sqrt{r^2 - (x-a)^2} + a^2 + b$ [2]

(c) By stating the values of a and b , solve the inequality $2x+1+\frac{1}{x-2} > \sqrt{16-(x-2)^2} + 5$. [3]

9 A curve R has equation $y = x^2 \ln x$ for $x > 0$. R crosses the x -axis at point A and has a turning point at B .

(a) State the coordinates of A . [1]

(b) Find the exact coordinates of B . [3]

(c) Find the exact area of the region bounded by R and the line segment AB . [5]

10 The functions f and g are defined by

$$f : x \mapsto 2x + 1, \quad x \in \mathbb{R},$$

$$g : x \mapsto \frac{ax-1}{bx-a}, \quad x \in \mathbb{R}, \quad x \neq \frac{a}{b},$$

where a and b are positive constants.

(a) Find $fg(1)$ in terms of a and b . You do not need to simplify your answer. [2]

(b) Explain why the function gf does not exist. [1]

(c) Find $g^{-1}(x)$. [2]

(d) Hence, or otherwise, find $g^2(x)$. [1]

(e) Given that $a > 1$ and $b = 1$, describe fully a sequence of transformations which transforms the curve

$y = g(x)$ onto the curve $y = \frac{1}{x}$. [4]

11 Ethylene is a gaseous plant hormone that plays an important role in inducing the ripening process for bananas. As a banana matures, ethylene is produced as a signal to induce fruit ripening. The amount of ethylene that a banana produces can be used to determine its ripeness.

Scientists set up an experiment to measure the amount of ethylene produced by a banana. The amount of ethylene, x parts per million (ppm), produced by a banana is modelled such that, at any time t hours, the rate of increase in the amount of ethylene is $k(160 - x)$, for some positive constant k .

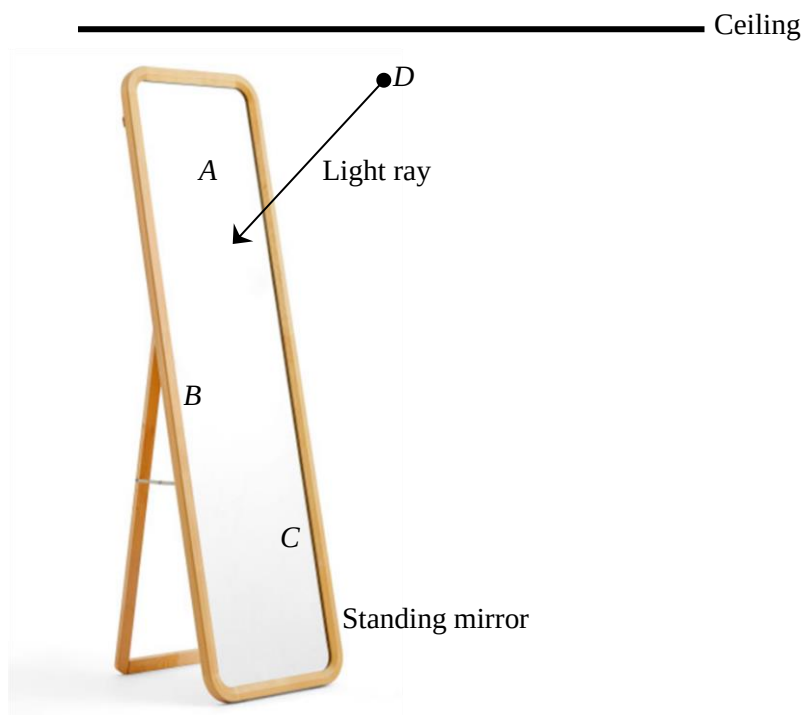
Initially no ethylene is present and it is found that there are 40 ppm of ethylene at time $t = 12$.

(a) Write down and solve a differential equation involving x and t . Find the time it takes for the amount of ethylene to reach 100 ppm. [4]

To reduce the amount of ethylene produced, ethylene absorbers can be used. The scientists repeat the experiment by adding ethylene absorbers at a rate of d ppm per hour. Initially, no ethylene is present and the rate of increase in the amount of ethylene, $k(160 - x)$, is the same as that found in part (a).

(b) Write down and solve a differential equation, giving x in terms of t and d . [6]

(c) Find the value of d for which the amount of ethylene will be kept at a maximum of 10 ppm in the long run. [2]



Credit: Image taken from <https://www.fortytwo.sg/theola-standing-mirror-birch.html>

In a room, there is a standing mirror and a light source at D below the ceiling. The three points A , B and C on the mirror, and D have position vectors $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = -\mathbf{j} + 6\mathbf{k}$, $\mathbf{c} = -\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$ and $\mathbf{d} = 5\mathbf{i} - \mathbf{j} + 8\mathbf{k}$ respectively with respect to a fixed point of reference.

(a) Show that the mirror can be modelled as a plane with cartesian equation $3x - 3y + 2z = 15$. [2]

(b) Find the exact position vector of point F , the foot of perpendicular from D to the mirror. [3]

A light ray is sent in the direction of $\begin{pmatrix} -6 \\ -2 \\ -13 \end{pmatrix}$ from D and hits the mirror at point E .

(c) Find the acute angle between the light ray and the mirror. [2]

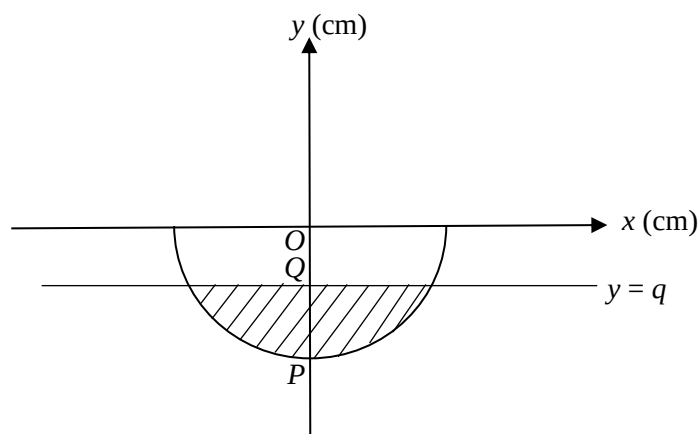
(d) Show that the coordinates of E is $\left(2, -2, \frac{3}{2}\right)$. [2]

(e) The light ray from D is reflected in the mirror. The reflected ray and the light ray lie in the same plane. The angle between the light ray and the mirror is the same as the angle between the reflected ray and the mirror. Find a vector equation of the reflected ray. [3]

Section A: Pure Mathematics [40 marks]

- 1 The area bounded by the loop of the curve $y^2 = (1-x)^2(x+3)$ is given by $2 \int_{\alpha}^{\beta} (1-x)\sqrt{x+3} \, dx$.
- (a) State the values of α and β . [1]
- (b) Use the substitution $u = \sqrt{x+3}$ to find the exact area bounded by this loop. [4]
- 2 With reference to the origin O , the points A , B and R have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{r}$ respectively. The points A and B are fixed and R varies.
- (a) Interpret geometrically the vector equation $\mathbf{r} = \lambda \mathbf{a}$ where $0 \leq \lambda \leq 1$. [1]
- (b) Interpret geometrically the vector equation $\mathbf{r} \times (\mathbf{a} - \mathbf{b}) = \mathbf{0}$. [2]
- (c) Interpret geometrically if $\mathbf{r} \cdot (\mathbf{r} - \mathbf{a}) = 0$. [2]
- (d) Given that $(\mathbf{r} - \mathbf{a}) \times (\mathbf{r} - \mathbf{b}) = \mathbf{0}$. Find $\mathbf{r}$ in terms of $\mathbf{a}$ and $\mathbf{b}$, showing your working clearly. [2]
- 3 **Do not use a calculator in answering the question.**
- Three complex numbers are $z_1 = -1 + \sqrt{3}i$, $z_2 = -1 - i$ and $z_3 = \sqrt{2} \left(\cos \frac{1}{12}\pi - i \sin \frac{1}{12}\pi \right)$.
- (a) Represent z_1 , z_2 and z_3 on an Argand diagram. [2]
- (b) Find $\frac{z_1}{z_2(z_3)^2}$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [4]
- A fourth complex number, z_4 , is such that $\frac{z_1 z_4}{z_2(z_3)^2}$ is purely real and $\left| \frac{z_1 z_4}{z_2(z_3)^2} \right| = 1$.
- (c) Find the possible values of z_4 in the form $r(\cos \theta + i \sin \theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$. [3]
- 4 (a) Write $\frac{1}{4r^2 - 8r + 3}$ in partial fractions. [2]
- (b) Find an expression in terms of n for $\sum_{r=2}^{3n} \left(\frac{1}{4r^2 - 8r + 3} \right)$. [3]
- (c) Find $\sum_{r=2}^{\infty} \left(\frac{1}{4r^2 - 8r + 3} \right)$. [1]
- (d) Using the result in part (b), find $\sum_{r=n+1}^{3n} \left(\frac{1}{4r^2 - 8r + 3} \right)$. Write your answer as a simplified fraction in terms of n . [3]

- 5 In this question units are in centimetres (cm).



The diagram shows part of the circle $x^2 + y^2 = 256$ and the line $y = q$, where $q < 0$. The circle and the line cross the negative y -axis at points P and Q respectively.

- (a) The shaded region between the circle and the line is rotated about the y -axis to form a wok of negligible thickness. The wok has depth PQ cm and a capacity of 3.3 litres. By finding the capacity of the wok in terms of q , calculate the depth of the wok giving your answer to the nearest integer.
[1 litre = 1000 cm³] [4]
- (b) A flat frying pan of negligible thickness with a capacity of 1.464π litres is formed by rotating the part of the circle between the lines $y = 0$ and $y = r$, $q < r < 0$, about the y -axis. Find a cubic equation satisfied by r and hence find the value of r . [3]
- (c) Water is poured from a container to the wok in part (a) at a rate of $\frac{3}{55}t$ litres per second, where t is the time in seconds from when pouring begins. Find the time taken to fill this wok to its full capacity of 3.3 litres. [3]

Section B: Probability and Statistics [60 marks]

- 6 A carpark has a row of 10 parking lots. On one day the carpark has 9 cars of different makes and 1 empty lot. Of the 9 cars, 2 are blue, 2 are red and the remaining 5 are of other distinct colours.
- (a) Find the number of different arrangements of the cars in the carpark. [1]
- (b) Find the number of different arrangements that can be made with both the blue cars next to each other and both the red cars next to each other. [2]
- (c) Find the number of different arrangements that can be made with no two adjacent cars the same colour. [3]
- 7 A computer randomly chooses 10 shapes from 18 squares and 12 triangles. The number of squares chosen is denoted by S .
- (a) The most probable number of squares that the computer chooses is denoted by s . By using the fact that $P(S = s) > P(S = s + 1)$, show that s satisfies the inequality

$$(s + 1)(s + 3) > (a - s)(b - s),$$

where a and b are constants to be determined. Hence find the value of s .

[4]

In a game, the computer randomly chooses 6 squares and 4 triangles. The squares are numbered from 1 to 6 and the triangles are numbered from 1 to 4. A square and a triangle are randomly chosen. Let X be the score which is defined as the difference (taken as always positive) between the numbers of the chosen square and triangle.

- (b) Find the probability distribution of X . [3]

Three independent games are played. The total score of the games is 3.

- (c) Find the probability that the score of the first game is 2. [3]

8 In this question, you should state the parameters of any distributions you use.

A baby quilt is made by stitching rectangular cotton fabric together. The length, in centimetres (cm), of each rectangular cotton fabric follows the distribution $N(24, 1.5^2)$ and the breadth follows the distribution $N(20, 1.2^2)$. Opposite sides of the rectangular cotton fabric are cut to the exact same measurement.

- (a) Find the probability that the length of a randomly chosen rectangular cotton fabric is less than 23.5 cm. [1]
- (b) Find the probability that the perimeter of a randomly chosen rectangular cotton fabric is more than 90 cm. [3]
- (c) State a necessary assumption for your calculations to hold in part (b). [1]
- (d) A baby quilt is made by stitching 48 pieces of rectangular cotton fabric together. Calculate the expected number of pieces that have length between 23 and 25 cm. [2]
- (e) Rectangular cotton fabrics can be paired together if their lengths differ from each other by at most k cm. Find the least value of k if 2 randomly chosen rectangular cotton fabrics have at least 90% chance of being paired together. [3]

9 A restaurant serves a signature dish. The probability that a diner orders the dish is 0.7. It may be assumed that each diner orders at most one portion of the dish and the number of diners who order the dish follows a binomial distribution.

- (a) Find the probability that 10 diners order at least 3 portions of the signature dish. [2]
- (b) Find the probability that 10 diners order between 3 and 8 portions of the signature dish. [2]
- (c) If the restaurant has enough ingredients to prepare 80 portions of the signature dish, find the maximum possible number of diners if the restaurant has at least 90% chance of meeting the number of orders for the dish. [2]

On a particular day, the restaurant caters for a function with 40 tables of 10 diners each.

- (d) Find the probability that each table orders at least 3 portions of the signature dish. [1]
- (e) Find an estimate of the probability that the average number of portions of the signature dish served per table is at least 6.9. [3]

- 10** A bus company is contracted to ferry students at a particular school on request. Following feedback from teachers and students that the buses usually arrive, on average, more than 15 minutes **after the scheduled pick-up time**, the school administration manager conducts a study of the arrival times of the bus after the scheduled pick-up time.

The administration manager collects a random sample of 30 bus arrival times after the scheduled pick-up time. Summary data for the arrival times after the scheduled pick-up time, t minutes, of these buses is as follows.

$$n = 30 \quad \sum t = 543 \quad \sum t^2 = 12722$$

- (a) Calculate unbiased estimates of the population mean and variance for the bus arrival time after the scheduled pick-up time. [2]
- (b) State hypotheses that can be used to test if the administration manager should agree with the feedback from teachers and students, defining any symbols you use. Work out the test statistic in this case, and use it to carry out the test at the 5% level of significance, giving your answer in the context of the question. [5]

The bus company makes changes to its operations and claims that their buses will arrive, on average, 15 minutes after the scheduled pick-up time. The administration manager then takes another sample of 40 bus arrival times after the scheduled pick-up time. He carries out a 2-tail test and finds no significant evidence to reject the company's claim at the 10% level of significance.

- (c) Use an algebraic method to evaluate the range of possible values of the sample mean bus arrival times after the scheduled pick-up time, $\bar{t}$. [3]
- (d) State two necessary assumptions for your calculations in part (c). [2]

- 11** Various studies suggest that plants exposed to music tend to grow taller compared to those not exposed to music. A junior biologist wants to investigate the effects of music on the growth of different plants. In one experiment, vines of a particular species were exposed to classical music. He recorded the average length of the vines, h centimetres, at age t days, in the table below.

t	7	10	13	16	19	22	25	28
h	19.2	25.4	36.5	41.4	45	k	51.9	53.8

- (a) Using the data, the junior biologist calculated the least squares regression line of h on t to be $h = 1.6393t + 11.475$. Show that the value of k is 48.1, correct to 1 decimal place. [2]

The junior biologist realised that the relationship between h and t is not linear and can be modelled by either

$$h = a\sqrt[3]{t} + b \text{ or } h = c\sqrt{t} + d,$$

where a , b , c and d are constants.

- (b) By calculating the relevant product moment correlation coefficients, explain why $h = a\sqrt[3]{t} + b$ is a better model than $h = c\sqrt{t} + d$. Find the equation of the better model. [4]

- (c) An estimate for the average length of the vines when they are 2 months old is obtained using the better model. Explain whether the estimate is reliable. [1]
- (d) Sketch a scatter diagram for h against $\sqrt[3]{t}$. Draw the line of best fit on your scatter diagram. [2]

For a line of best fit $y = f(x)$, the residual is the difference between the observed and predicted value. The difference between an observed value (p, q) plotted on the scatter diagram and the predicted value $(p, f(p))$ is defined as $q - f(p)$.

- (e) Find the value of the residual corresponding to $t = 28$, for the line of best fit found in part (b). [1]
- (f) Explain why the line of best fit is referred to as the ‘least squares’ regression line in relation to the residuals. [2]

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Q1	Suggested Answers	Guidance
	Make w the subject from $w + i + 3 = v$: $w = v - i - 3$ Sub into $v^2 - iw + 2 = 0$ $v^2 - i(v - i - 3) + 2 = 0$ $v^2 - iv + 1 + 3i = 0$ $v = \frac{i \pm \sqrt{-1 - 4(1 + 3i)}}{2}$ $= \frac{i \pm \sqrt{-5 - 12i}}{2}$ $= \frac{i \pm (2 - 3i)}{2}$ $= \frac{i + (2 - 3i)}{2} \text{ or } \frac{i - (2 - 3i)}{2}$ $= 1 - i \text{ or } -1 + 2i$ $w = -2 - 2i \text{ or } -4 + i$	Generally, most students can rewrite as a quadratic equation in terms of v or w. The recommended method to solve a quadratic equation with imaginary coefficient(s) is to use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Can use GC to get $\sqrt{-5 - 12i}$.
	Alternative (1) Sub $v = w + i + 3$ into $v^2 - iw + 2 = 0$ $(w + i + 3)^2 - iw + 2 = 0$ $w^2 + 2(i + 3)w + (i + 3)^2 - iw + 2 = 0$ $w^2 + (i + 6)w + (10 + 6i) = 0$ $w = \frac{-(6 + i) \pm \sqrt{(6 + i)^2 - 4(10 + 6i)}}{2}$	

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	$= \frac{-(6+i) \pm \sqrt{-5-12i}}{2} = \frac{-(6+i) \pm (2-3i)}{2}$ $w = -2 - 2i, -4 + i$ Using $v = w + i + 3$ We have $v = 1 - i, -1 + 2i$	
	Alternative (2) From the equation $w + i + 3 = v$ Multiply by i, we have $iw - 1 + 3i = iv \dots (1)$ $v^2 - iw + 2 = 0 \dots (2)$ (1)+(2): $v^2 - iv + 1 + 3i = 0$ $v = \frac{i \pm \sqrt{-1-4(1+3i)}}{2}$ $= \frac{i \pm \sqrt{-5-12i}}{2}$ $= \frac{i \pm (2-3i)}{2}$ $= \frac{i + (2-3i)}{2} \text{ or } \frac{i - (2-3i)}{2}$ $= 1 - i \text{ or } -1 + 2i$ $w = -2 - 2i \text{ or } -4 + i$	

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Q2	Suggested Answers	Guidance
(a)	$x = a\theta - a \sin \theta$ $\frac{dx}{d\theta} = a - a \cos \theta$ $y = a - a \cos \theta$ $\frac{dy}{d\theta} = a \sin \theta$ $\frac{dy}{dx} = \frac{a \sin \theta}{a(1 - \cos \theta)}$ $= \frac{\left(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}\right)}{\left(1 - \left(1 - 2 \sin^2 \frac{\theta}{2}\right)\right)}$ $= \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} = \cot \frac{\theta}{2}$	For show part in the question, clear workings must be written, in this case, the use of double angle formula for $\frac{\left(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}\right)}{\left(1 - \left(1 - 2 \sin^2 \frac{\theta}{2}\right)\right)}$ must be explicitly written.
(b)	When $\theta = \frac{\pi}{3}$, $x = \frac{\pi}{3}a - \frac{\sqrt{3}}{2}a$, $y = \frac{1}{2}a$ and $\frac{dy}{dx} = \cot \frac{\pi}{6} = \sqrt{3}$ Equation of tangent is $y - \frac{1}{2}a = \sqrt{3}\left(x - \frac{\pi}{3}a + \frac{\sqrt{3}}{2}a\right)$	When $\theta = \frac{\pi}{3}$, x and y coordinates, with gradient $\cot \frac{\pi}{6} = \sqrt{3}$ must be stated clearly.

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$\therefore y = \sqrt{3}x - \frac{\sqrt{3}}{3}\pi a + 2a$ When $x = \frac{\pi}{3}a$, $y = \sqrt{3}\left(\frac{\pi}{3}a\right) - \frac{\sqrt{3}}{3}\pi a + 2a = 2a$ Therefore, the tangent passes through $\left(\frac{1}{3}\pi a, 2a\right)$ Or When $y = 2a$, $2a = \sqrt{3}x - \frac{\sqrt{3}}{3}\pi a + 2a \Rightarrow x = \frac{\frac{\sqrt{3}}{3}\pi a}{\sqrt{3}} = \frac{1}{3}\pi a$ Therefore, the tangent passes through $\left(\frac{1}{3}\pi a, 2a\right)$	Once again, this is a show part, all working must be explicitly written.
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Q3	Suggested Answers	Guidance
(a)	$y = (\sin^{-1} x)^2$ Differentiate wrt x: $\frac{dy}{dx} = 2(\sin^{-1} x) \frac{1}{\sqrt{1-x^2}}$ $\sqrt{1-x^2} \frac{dy}{dx} = 2 \sin^{-1} x$ Differentiate wrt x: $\sqrt{1-x^2} \frac{d^2y}{dx^2} - \frac{x}{\sqrt{1-x^2}} \frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$ $(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 2$ Alternatively, $\sqrt{1-x^2} \frac{dy}{dx} = 2 \sin^{-1} x$ $\left(\sqrt{1-x^2} \frac{dy}{dx} \right)^2 = (2 \sin^{-1} x)^2$ $(1-x^2) \left(\frac{dy}{dx} \right)^2 = 4y$ Differentiate wrt x: $(1-x^2) \cdot 2 \frac{dy}{dx} \frac{d^2y}{dx^2} - 2x \left(\frac{dy}{dx} \right)^2 = 4 \frac{dy}{dx}$ $(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 2$ $(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - 2 = 0$	When such questions ask the student to show a differential equation $(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - 2 = 0$, one simply has to differentiate and simplify to get the required equation. There is no need to find the derivatives $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and substitute into the equation to check that LHS = RHS. We do not need to make y, $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ the subject before differentiating. Just do implicit differentiation.

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(b)	Differentiate wrt x : $(1-x^2)\frac{d^3y}{dx^3}-2x\frac{d^2y}{dx^2}-\left(x\frac{d^2y}{dx^2}+\frac{dy}{dx}\right)=0$ $(1-x^2)\frac{d^3y}{dx^3}-3x\frac{d^2y}{dx^2}-\frac{dy}{dx}=0$ Differentiate wrt x : $(1-x^2)\frac{d^4y}{dx^4}-2x\frac{d^3y}{dx^3}-3\left(x\frac{d^3y}{dx^3}+\frac{d^2y}{dx^2}\right)-\frac{d^2y}{dx^2}=0$ $(1-x^2)\frac{d^4y}{dx^4}-5x\frac{d^3y}{dx^3}-4\frac{d^2y}{dx^2}=0$ When $x=0$, $y=0$, $\frac{dy}{dx}=0$, $\frac{d^2y}{dx^2}=2$, $\frac{d^3y}{dx^3}=0$ and $\frac{d^4y}{dx^4}=8$ $\therefore y=2\left(\frac{x^2}{2!}\right)+8\left(\frac{x^4}{4!}\right)+\dots$ $y=x^2+\frac{1}{3}x^4+\dots$	
(c)	$y \approx x^2 + \frac{1}{3}x^4$ $(\sin^{-1} x)^2 \approx x^2 + \frac{1}{3}x^4$ When $x = \frac{1}{2}$, $\left(\sin^{-1} \frac{1}{2}\right)^2 \approx \left(\frac{1}{2}\right)^2 + \frac{1}{3}\left(\frac{1}{2}\right)^4$ $\frac{\pi^2}{36} \approx \frac{13}{48}$	Many students are unable to find an approximation for π . When we find the Maclaurin series for y, it is important to take note (and write down) what y is. In this case $y = (\sin^{-1} x)^2$ and hence from the Maclaurin series, we have $(\sin^{-1} x)^2 = x^2 + \frac{1}{3}x^4 + \dots$

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	$\pi \approx \frac{\sqrt{39}}{2}$ Percentage error of approximation = $\frac{\pi - \frac{\sqrt{39}}{2}}{\pi} \times 100\%$ $= 0.608\% \quad (3 \text{ s.f.})$ Approximation is quite accurate as the value of x is sufficiently small and the number of terms up to power 4 used is sufficient.	To comment on the accuracy of the approximation, we first make a stand that the approximation is accurate (percentage error is small). Then we provide the justification. Usually, there are 2 related factors that we consider: (1) x is small so that terms with higher powers are negligible, (2) terms with sufficiently high powers of x are included. (To improve accuracy, we use a smaller value of x when the number of terms are fixed, and we use more terms when the value of x is fixed.) Usually x is small means 0.1, 0.01, etc. Here, $x = \frac{1}{2}$ is not quite small, and so it is necessary to mention that sufficient number of terms is used to explain why the approximation is quite accurate.
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Q4	Suggested Answers	Guidance
(a)	$f(x) = \sqrt{3} \cos x + \sin x = R \cos(x - \alpha)$ $R = \sqrt{(\sqrt{3})^2 + 1^2} = 2$ $\alpha = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$	If you forget the method, you can derive: Use the formula in MF26 to expand $R \cos(x - \alpha) =$ $R(\cos x \cos \alpha + \sin x \sin \alpha)$ Then compare the coefficients of $\cos x$ and $\sin x$ to get $R \cos \alpha = \sqrt{3}, \quad R \sin \alpha = 1$
(b)	$\int_0^{\frac{\pi}{6}} \left(\frac{1}{f(x)} \right)^2 dx$ $= \int_0^{\frac{\pi}{6}} \frac{1}{\left(2 \cos \left(x - \frac{\pi}{6} \right) \right)^2} dx$ $= \frac{1}{4} \int_0^{\frac{\pi}{6}} \sec^2 \left(x - \frac{\pi}{6} \right) dx$ $= \frac{1}{4} \left[\tan \left(x - \frac{\pi}{6} \right) \right]_0^{\frac{\pi}{6}}$ $= \frac{1}{4} \left[\tan 0 - \tan \left(-\frac{\pi}{6} \right) \right]$ $= \frac{1}{4\sqrt{3}} = \frac{\sqrt{3}}{12}$	

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(c)	$ \begin{aligned} & \int_0^{\frac{\pi}{12}} \frac{1}{f(2x)} dx \\ &= \int_0^{\frac{\pi}{12}} \frac{1}{2 \cos\left(2x - \frac{\pi}{6}\right)} dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{12}} \sec\left(2x - \frac{\pi}{6}\right) dx \\ &= \frac{1}{4} \int_0^{\frac{\pi}{12}} 2 \sec\left(2x - \frac{\pi}{6}\right) dx \\ &= \frac{1}{4} \ln \left[\left \sec\left(2x - \frac{\pi}{6}\right) + \tan\left(2x - \frac{\pi}{6}\right) \right \right]_0^{\frac{\pi}{12}} \\ &= \frac{1}{4} \left[\ln \sec 0 + \tan 0 - \ln \left \sec\left(-\frac{\pi}{6}\right) + \tan\left(-\frac{\pi}{6}\right) \right \right] \\ &= \frac{1}{4} \left[\ln 1 - \ln \left \frac{2}{\sqrt{3}} + \left(-\frac{2}{\sqrt{3}}\right) \right \right] \\ &= -\frac{1}{4} \ln \frac{1}{\sqrt{3}} \\ &= \frac{1}{4} \ln \sqrt{3} \\ &= \frac{1}{8} \ln 3 \end{aligned} $	
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Q5	Suggested Answers	Guidance
(a)	Method 1: $\int \frac{4x-1}{x^2+4x+4} dx$ $= \int \frac{2(2x+4)-9}{x^2+4x+4} dx$ $= 2 \int \frac{2x+4}{x^2+4x+4} dx - \int \frac{9}{(x+2)^2} dx$ $= 2 \ln(x^2+4x+4) + \frac{9}{x+2} + C$ $= 4 \ln(x+2) + \frac{9}{x+2} + C$ Method 2: Use of partial fractions $\frac{4x-1}{x^2+4x+4} = \frac{4x-1}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2}$ $4x-1 = A(x+2) + B$ Solving, $A = 4$, $B = -9$ $\int \frac{4x-1}{x^2+4x+4} dx = \int \frac{4}{x+2} - \frac{9}{(x+2)^2} dx$ $= 4 \ln(x+2) + \frac{9}{x+2} + C$	Final answer must be in simplest form. Complete the square for $x^2 + 4x + 4$ $2 \ln(x^2 + 4x + 4)$ $= 2 \ln(x+2)^2$ $= 4 \ln(x+2)$
(b)	$\int_0^1 \frac{ 4x-1 }{x^2+4x+4} dx$ $= -\int_0^{\frac{1}{4}} \frac{4x-1}{x^2+4x+4} dx + \int_{\frac{1}{4}}^1 \frac{4x-1}{x^2+4x+4} dx$	$ 4x-1 = \begin{cases} -(4x-1), & x < \frac{1}{4} \\ 4x-1, & x \geq \frac{1}{4} \end{cases}$

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	$= - \left[4 \ln(x+2) + \frac{9}{x+2} \right]_{\frac{1}{4}}^{\frac{1}{4}} + \left[4 \ln(x+2) + \frac{9}{x+2} \right]_{\frac{1}{4}}^1$ $= - \left[\left(4 \ln \frac{9}{4} + 4 \right) - \left(4 \ln 2 + \frac{9}{2} \right) \right] + \left[(4 \ln 3 + 3) - \left(4 \ln \frac{9}{4} + 4 \right) \right]$ $= -4 \ln \frac{9}{4} + 4 \ln 2 + 4 \ln 3 - 4 \ln \frac{9}{4} - \frac{1}{2}$ $= 4 \ln \frac{2 \times 3}{\left(\frac{9}{4} \times \frac{9}{4} \right)} - \frac{1}{2} = 4 \ln \frac{32}{27} - \frac{1}{2}$	
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Q6	Suggested Answers	Guidance						
(a)	$\frac{a+5d}{a} = \frac{a+13d}{a+5d}$ $(a+5d)^2 = a(a+13d)$ $a^2 + 10ad + 25d^2 = a^2 + 13ad$ $25d^2 = 3ad$ Since $d \neq 0$, $3a = 25d$ $d = \frac{3a}{25}$							
(b)(i)	Sum to infinity = $\frac{b}{1-0.5} = 2b$	Apply formula for sum to infinity $S_{\infty} = \frac{a}{1-r}$						
(ii)	$S_n = \frac{b(1-0.5^n)}{1-0.5} = 2b(1-0.5^n)$ $S_{2n} - S_n < 0.004b \quad \text{or} \quad S_n - S_{2n} < 0.004b$ $2b(1-0.5^{2n}) - 2b(1-0.5^n) < 0.004b$ $0.5^n - 0.5^{2n} < 0.002, \text{ since } b > 0$ $0.5^n - 0.5^{2n} - 0.002 < 0$ Using GC, <table><tr><td>n</td><td>$0.5^n - 0.5^{2n} - 0.002$</td></tr><tr><td>8</td><td>$0.00189 > 0$</td></tr><tr><td>9</td><td>$-5.07 \times 10^{-5} < 0$</td></tr></table> Smallest $n = 9$	n	$0.5^n - 0.5^{2n} - 0.002$	8	$0.00189 > 0$	9	$-5.07 \times 10^{-5} < 0$	Since $S_n < S_{2n}$, we use either $S_{2n} - S_n$ or $S_n - S_{2n}$. Modulus sign must be present in the 2nd expression. When solving $S_n - S_{2n} < 0.004b$, there is no need for removal of modulus sign. Key the expression $2(1-0.5^n) - 2(1-0.5^{2n}) - 0.004$ into the GC to get required answer. Show the table of values as your working.
n	$0.5^n - 0.5^{2n} - 0.002$							
8	$0.00189 > 0$							
9	$-5.07 \times 10^{-5} < 0$							

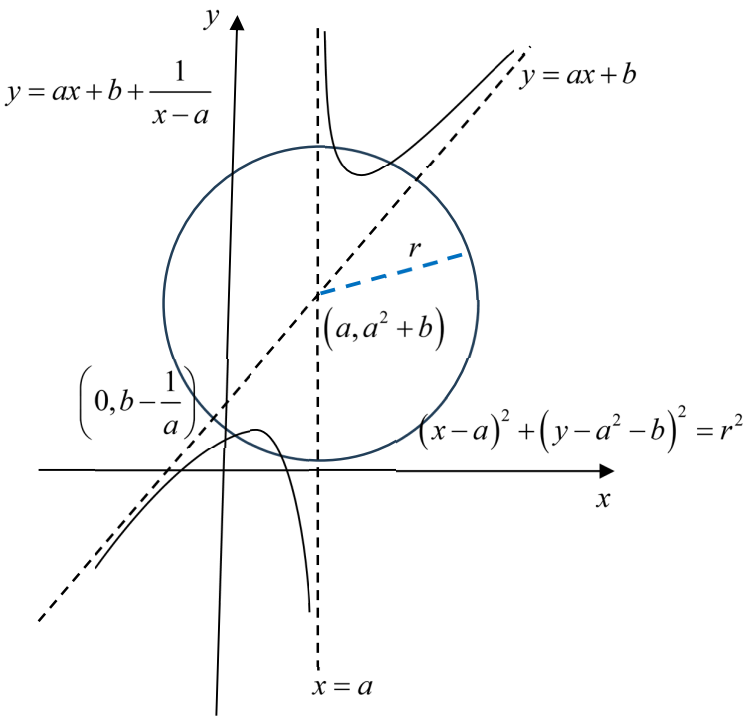
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Q7	Suggested Answers	Guidance
	Length of height = $1 + \cos \theta$ and length of base = $2 \sin \theta$ Area of triangle, $S = \frac{1}{2} \times (1 + \cos \theta) \times 2 \sin \theta$ $= \sin \theta + \frac{1}{2} \sin 2\theta$ Alternatively, Area of triangle, $S = 2 \left(\frac{1}{2} \times (1)^2 \sin(\pi - \theta) \right) + \frac{1}{2} \times (1)^2 \sin 2\theta$ $= \sin \theta + \frac{1}{2} \sin 2\theta$ $\frac{dS}{d\theta} = \cos \theta + \cos 2\theta$ For maximum area, $\frac{dS}{d\theta} = 0$, ie, $\cos \theta + \cos 2\theta = 0$ $2 \cos^2 \theta + \cos \theta - 1 = 0$ $(2 \cos \theta - 1)(\cos \theta + 1) = 0$ $\cos \theta = \frac{1}{2} \text{ or } \cos \theta = -1 \text{ (rejected since } \theta \text{ is acute)}$ Therefore, $\theta = \frac{\pi}{3}$	An appropriate diagram would have helped identify the correct length and height of the isosceles triangle. Some students used an alternative method by finding the length AB or AC. However, they quoted the wrong cosine rule and were not successful in finding the required length. As this is a “show” question, students need to show sufficient working to deduce that the final acceptable angle for θ is $\frac{\pi}{3}$. However, with the correct value of $\theta = \frac{\pi}{3}$, it is insufficient to imply that the maximum area of triangle ABC is equilateral as θ is not the angle in the triangle unless explained clearly. It is also incorrect to assume that the angle is $\frac{\pi}{3}$ and proceed to use this to show that the area of triangle ABC is maximum when it is equilateral. The negative value of the second derivative when $\theta = \frac{\pi}{3}$ needs to be seen when proving for maximum.

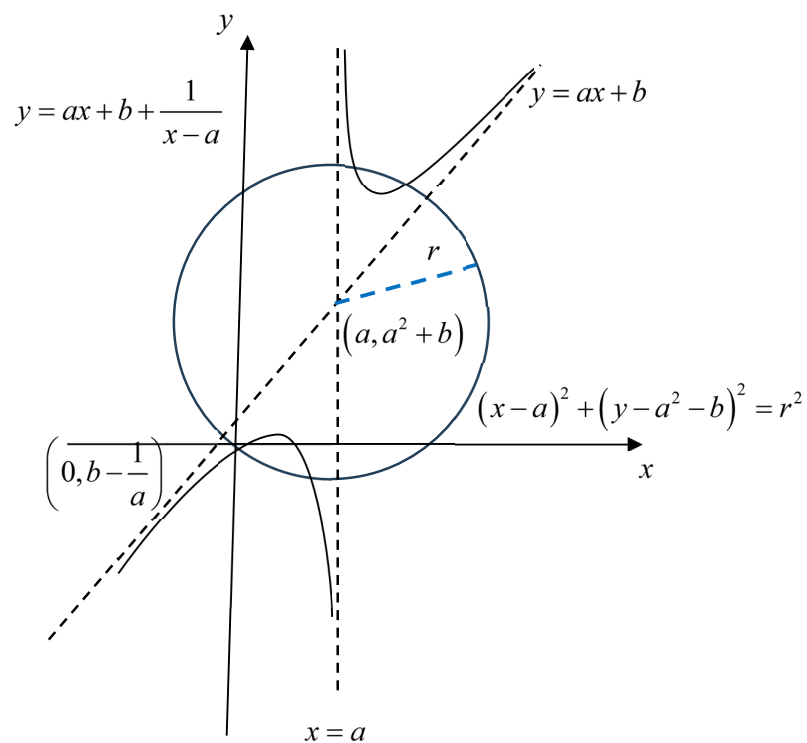
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$\frac{d^2S}{d\theta^2} = -\sin\theta - 2\sin 2\theta$ When $\theta = \frac{\pi}{3}$, $\frac{d^2S}{d\theta^2} = -\frac{3\sqrt{3}}{2} < 0$ Since $\angle BOC = 2\theta$, $\angle BAC = \theta = \frac{\pi}{3}$. As triangle is isosceles, all the angles in the triangle are $\frac{\pi}{3}$. Therefore, maximum area occurs when triangle ABC is equilateral, ie, when $\theta = \frac{\pi}{3}$ Maximum area $= \sin\frac{\pi}{3} + \frac{1}{2}\sin\frac{2\pi}{3} = \frac{3}{4}\sqrt{3}$	Many missed the word ‘exactly’ and left the answer to 3 sf.
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Q8	Suggested Answers	Guidance
(a) (b)	y-intercept $b - \frac{1}{a} > 0$  The graph shows a Cartesian coordinate system with x and y axes. A circle is drawn with its center at the point $(a, a^2 + b)$. A dashed line represents the asymptote $y = ax + b$. Another dashed vertical line represents the asymptote $x = a$. The circle passes through the point $(0, b - \frac{1}{a})$ on the y-axis. The equation of the circle is given as $(x - a)^2 + (y - a^2 - b)^2 = r^2$. The radius r is indicated by a dashed line from the center to the circle's edge.	Coordinates of the points where the curve crosses the x-axis are not required. Note that the centre of the circle must be clearly indicated on your sketch, and it is exactly at the point of intersection of the 2 asymptotes. Maximum point of the hyperbola is on the right of the y-intercept of the hyperbola and it must be drawn clearly. Circles must be drawn using a compass.

Or y -intercept $b - \frac{1}{a} < 0$



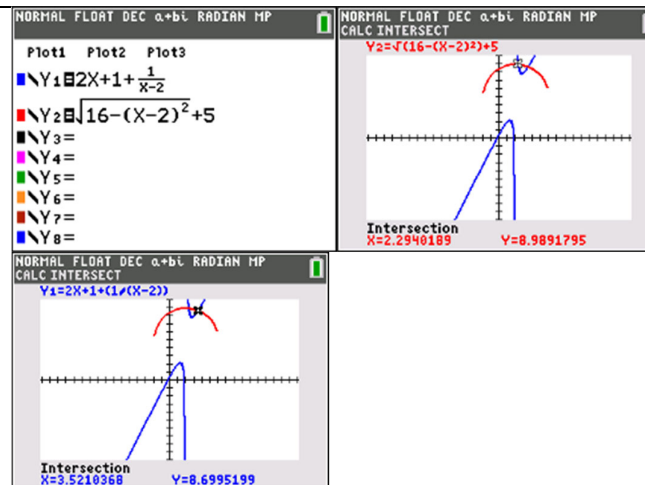
(c)

$a = 2$ and $b = 1$

Using GC,
the x -coordinates of the points of intersection are 2.29 and 3.52

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$$2 < x < 2.29 \text{ or } 3.52 < x \leq 6$$



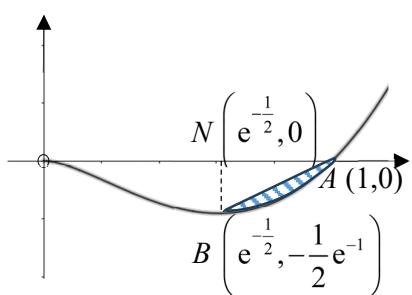
$$2x + 1 + \frac{1}{x-2}$$

$$= \sqrt{16 - (x-2)^2} + 5$$

shows the intersection of the hyperbola and upper part of the **circle with centre (2,5) and radius 4**. The value of x cannot exceed 6 for

$\sqrt{16 - (x-2)^2}$ to be defined.

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Q9	Suggested Answers	Guidance
(a)	Coordinates of A are $(1,0)$	Note that question states to find coordinates and $x > 0$.
(b)	$y = x^2 \ln x$ $\frac{dy}{dx} = 2x \ln x + x^2 \left(\frac{1}{x}\right) = x(2 \ln x + 1)$ $x(2 \ln x + 1) = 0$ $x = 0$ or $\ln x = -\frac{1}{2}$ Since $x > 0$, $x = e^{-\frac{1}{2}}$ Coordinates of B are $\left(e^{-\frac{1}{2}}, -\frac{1}{2}e^{-1}\right)$	Use of brackets advised so as not to misread the simplification of $\left(e^{-\frac{1}{2}}\right)^2 \ln\left(e^{-\frac{1}{2}}\right) = e^{-1}\left(-\frac{1}{2}\right)$ as $e^{-1} - \frac{1}{2}$.
(c)	 Method 1: Area required $= \left(-\int_{e^{-\frac{1}{2}}}^1 (x^2 \ln x) dx\right) - \text{area of triangle } ABN$ $= -\left(\left[\frac{1}{3}x^3 \ln x\right]_{e^{-\frac{1}{2}}}^1 - \frac{1}{3}\int_{e^{-\frac{1}{2}}}^1 x^2 dx\right) - \frac{1}{4}e^{-1}\left(1 - e^{-\frac{1}{2}}\right)$ $u = \ln x \quad \frac{dv}{dx} = x^2$ $\frac{du}{dx} = \frac{1}{x} \quad v = \frac{1}{3}x^3$	It is strongly advised to have a sketch of the curve $y = x^2 \ln x$ and the line segment AB to identify correctly the region required. Many attempted to find the area of triangle but using the wrong height $-\frac{1}{2}e^{-1}$ instead of $\frac{1}{2}e^{-1}$. Other common mistakes include failure to consider the correct integral as region bounded by the curve and x-axis is below the x-axis and careless algebraic manipulations when negative signs are present. It should also be noted that answers should be simplified where possible.

$= - \left(\left[0 - \frac{1}{3} e^{-\frac{3}{2}} \left(-\frac{1}{2} \right) \right] - \frac{1}{9} [x^3]_{e^{-\frac{1}{2}}}^1 \right) - \frac{1}{4} e^{-1} \left(1 - e^{-\frac{1}{2}} \right)$ $= -\frac{1}{6} e^{-\frac{3}{2}} + \frac{1}{9} \left(1 - e^{-\frac{3}{2}} \right) - \frac{1}{4} e^{-1} \left(1 - e^{-\frac{1}{2}} \right)$ $= \frac{1}{9} - \frac{1}{36} e^{-\frac{3}{2}} - \frac{1}{4} e^{-1}$	
Method 2: (area bounded between line and curve) Equation of line joining A and B is $y = \frac{\frac{1}{2} e^{-1}}{1 - e^{-\frac{1}{2}}} (x - 1) = \frac{1}{2e(1 - e^{-\frac{1}{2}})} (x - 1) = \frac{1}{2(e - e^{\frac{1}{2}})} (x - 1)$ Area required $= \int_{e^{-\frac{1}{2}}}^1 \frac{1}{2(e - e^{\frac{1}{2}})} (x - 1) dx - \int_{e^{-\frac{1}{2}}}^1 (x^2 \ln x) dx$ $= \frac{1}{2(e - e^{\frac{1}{2}})} \left[\frac{x^2}{2} - x \right]_{e^{-\frac{1}{2}}}^1 - \left\{ \left[\frac{1}{3} x^3 \ln x \right]_{e^{-\frac{1}{2}}}^1 - \frac{1}{3} \int_{e^{-\frac{1}{2}}}^1 x^2 dx \right\}$ $= \frac{1}{2(e - e^{\frac{1}{2}})} \left[\left(\frac{1}{2} - 1 \right) - \left(\frac{e^{-1}}{2} - e^{-\frac{1}{2}} \right) \right] - \left\{ \left[0 - \frac{1}{3} e^{-\frac{3}{2}} \left(-\frac{1}{2} \right) \right] - \frac{1}{9} [x^3]_{e^{-\frac{1}{2}}}^1 \right\}$ $= \frac{1}{2(e - e^{\frac{1}{2}})} \left[e^{-\frac{1}{2}} - \frac{1}{2} - \frac{e^{-1}}{2} \right] - \frac{1}{6} e^{-\frac{3}{2}} + \frac{1}{9} \left(1 - e^{-\frac{3}{2}} \right)$	

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	$= \frac{1}{4(e - e^{\frac{1}{2}})} \left[2e^{-\frac{1}{2}} - 1 - e^{-1} \right] - \frac{5}{18} e^{-\frac{3}{2}} + \frac{1}{9}$	
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Q10	Suggested Answers	Guidance
(a)	$fg(1) = f\left(\frac{a-1}{b-a}\right)$ $= 2\left(\frac{a-1}{b-a}\right) + 1$	
(b)	$R_f = \mathbb{R} \text{ and } D_g = \mathbb{R} \setminus \left\{\frac{a}{b}\right\}$ $\frac{a}{b} \in R_f \text{ but } \frac{a}{b} \notin D_g$ Therefore, gf does not exist.	As this is a given answer, stating the condition that the range of the function f is not a subset of the domain of the function g provides insufficient detail to answer the question fully. To show that the composite does not exist, it is necessary to identify that a specific value of $\frac{a}{b}$ is in the range of f but not in the domain of g, so that the function gf does not exist <u>as required by the exam board</u>. It is necessary to write in proper notation for range and domain of functions.
(c)	Let $y = \frac{ax-1}{bx-a}$ $bxy - ay = ax - 1$ $bxy - ax = ay - 1$ $x = \frac{ay-1}{by-a}$ $g^{-1}(x) = \frac{ax-1}{bx-a}$	
(d)	Hence method: Since $g(x) = g^{-1}(x)$ $gg(x) = gg^{-1}(x)$ $g^2(x) = x$	

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	Otherwise method: $g^2(x) = gg(x)$ $= g\left(\frac{ax-1}{bx-a}\right)$ $= \frac{a\left(\frac{ax-1}{bx-a}\right)-1}{b\left(\frac{ax-1}{bx-a}\right)-a}$ $= \frac{a(ax-1)-(bx-a)}{b(ax-1)-a(bx-a)}$ $= \frac{a^2x-a-bx+a}{abx-b-abx+a^2}$ $= \frac{a^2x-bx}{a^2-b}$ $= x$	The otherwise method is not recommended as it is too long for a 1 mark question.
(e)	$g(x) = \frac{ax-1}{x-a} = a + \frac{a^2-1}{x-a}$ (1) Translate the graph of $y = g(x)$ by $-a$ units in the direction of the y-axis. (2) Scaling by a factor of $\frac{1}{a^2-1}$ parallel to the y-axis. (3) Translate $-a$ units in the direction of the x-axis. Note: For this qn, the following order are accepted also (1), (3), (2) (3), (1), (2)	Transformation is easier to describe when $g(x)$ is expressed in the form of a proper fraction. For description of transformations, the correct phrasing must be seen. For translation, some students wrote “along the axis” which is not accepted as it does not give a specific direction. For scaling, some students wrote “scale by __ units”, which is not accepted. Many students referred to the graph of $y = g(x)$ for the 2nd and 3rd transformations, which is incorrect as the curve would take a new equation after the 1st transformation is done.

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	Transformation (1) must come before (2) if the above descriptions are used.	
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Q11	Suggested Answers	Guidance
(a)	$\frac{dx}{dt} = k(160 - x)$ $\frac{1}{160 - x} \frac{dx}{dt} = k$ $\int \frac{1}{160 - x} dx = k \int 1 dt$ $-\ln 160 - x = kt + C$ $160 - x = Ae^{-kt} \quad \text{where } A = \pm e^{-C}$ $t = 0, x = 0 \Rightarrow A = 160$ $160 - x = 160e^{-kt}$ $t = 12, x = 40 \Rightarrow 120 = 160e^{-12k}$ $\frac{3}{4} = e^{-12k}$ $k = -\frac{1}{12} \ln \frac{3}{4}$ $160 - x = 160e^{\left(\frac{1}{12} \ln \frac{3}{4}\right)t} = 160 \left(\frac{3}{4}\right)^{\frac{t}{12}}$ $x = 100 \Rightarrow 60 = 160 \left(\frac{3}{4}\right)^{\frac{t}{12}}$ $t = 40.9 \text{ (3 s.f.)}$	Generally well done by most candidates. Some candidates substituted in the initial conditions to solve for k and C before removing the $\ln$ and modulus. If this method is used, candidates will end up with the modulus in the $\ln$ function, which they will have to determine which solution to be rejected or accepted. This process is more tedious and should be avoided.
(b)	$\frac{dx}{dt} = k(160 - x) - d = (160k - d) - kx$ $\frac{1}{(160k - d) - kx} \frac{dx}{dt} = 1$	The tip here is to recognize that x is the only variable here, and k & d are constants to separate the variables correctly. Candidates should realise that the value of k from (a) is to be used. To keep working tidy, candidates should avoid substituting in k until after making x the subject.

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	$-\frac{1}{k} \ln (160k - d) - kx = t + C$ $\ln (160k - d) - kx = -kt - kC$ $(160k - d) - kx = Be^{-kt} \quad \text{where } B = \pm e^{-kC}$ $t = 0, x = 0 \Rightarrow B = 160k - d$ $(160k - d) - kx = (160k - d)e^{-kt}$ $kx = (160k - d)(1 - e^{-kt})$ $x = \left(160 - \frac{d}{k}\right)(1 - e^{-kt})$ $= \left(160 + \frac{12d}{\ln \frac{3}{4}}\right) \left(1 - e^{\left(\frac{1}{12} \ln \frac{3}{4}\right)t}\right) = \left(160 + \frac{12d}{\ln \frac{3}{4}}\right) \left(1 - \left(\frac{3}{4}\right)^{\frac{t}{12}}\right)$	
(c)	As $t \rightarrow \infty, \left(\frac{3}{4}\right)^{\frac{t}{12}} \rightarrow 0$ $\Rightarrow x \rightarrow 160 + \frac{12d}{\ln \frac{3}{4}}$ $160 + \frac{12d}{\ln \frac{3}{4}} = 10$ $d = 3.5960 \text{ (5 s.f.)} = 3.60 \text{ (3 s.f.)}$	“In the long run” suggests that candidates should obtain the asymptotic value of x as $t \rightarrow \infty$

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Q12	Suggested Answers	Guidance
(a)	$\overrightarrow{AB} = \begin{pmatrix} -2 \\ 0 \\ 3 \end{pmatrix} \quad \overrightarrow{BC} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} \quad \overrightarrow{AC} = \begin{pmatrix} -3 \\ -1 \\ 3 \end{pmatrix}$ $\overrightarrow{AB} \times \overrightarrow{BC} = \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix}$ $\text{Equation of plane } ABC : \mathbf{r} \cdot \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} = 15$ $3x - 3y + 2z = 15$	This being a “show” question, student should explain clearly how the normal vector $\begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix}$ is obtained.
(b)	$\mathbf{r} = \begin{pmatrix} 5 \\ -1 \\ 8 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix}, \mu \in \mathbb{R}$ $\text{Equation of plane } ABC : \mathbf{r} \cdot \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} = 15$ At intersection, $\begin{pmatrix} 5+3\mu \\ -1-3\mu \\ 8+2\mu \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} = 15$ $34 + 22\mu = 15 \Rightarrow \mu = -\frac{19}{22}$	The solution shown here is the most common method used. Common mistakes made: - Students wrote $\begin{pmatrix} 5+3\mu \\ -1-3\mu \\ 8+2\mu \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} = 0$ - Thought that equation of line DF: $\mathbf{r} = \begin{pmatrix} 5 \\ -1 \\ 8 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix}$ is $\overrightarrow{DF}$.

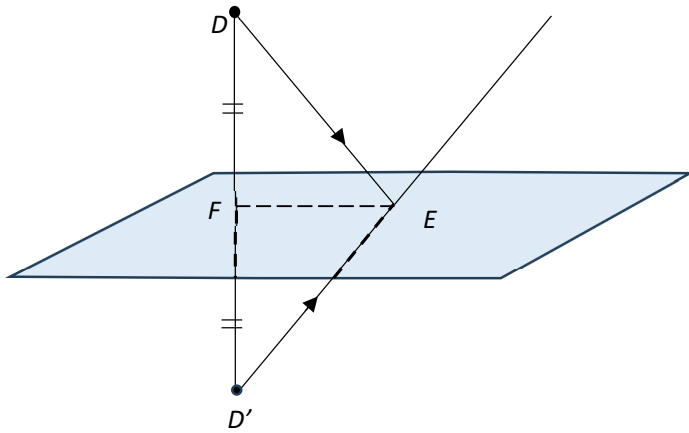
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	Position of the foot of perpendicular is $\overrightarrow{OF} = \frac{1}{22} \begin{pmatrix} 53 \\ 35 \\ 138 \end{pmatrix}$	
(c)	Let θ be the angle between the light ray and the mirror. $\sin \theta = \frac{\left \begin{pmatrix} -6 \\ -2 \\ -13 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} \right }{\left \begin{pmatrix} -6 \\ -2 \\ -13 \end{pmatrix} \right \left \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} \right }$ $\sin \theta = \frac{ -38 }{\sqrt{209} \times 22} \Rightarrow \theta = 34.1^\circ \quad (\text{or } 0.595 \text{ radian})$ Alternative: Let ϕ be the angle between the light ray and the normal of the mirror. $\text{Then } \cos \phi = \frac{\left \begin{pmatrix} -6 \\ -2 \\ -13 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} \right }{\left \begin{pmatrix} -6 \\ -2 \\ -13 \end{pmatrix} \right \left \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} \right }$ $\phi = 55.9^\circ$ Hence angle between the mirror and light ray $= 90^\circ - 55.9^\circ = 34.1^\circ$	This part is generally well-attempted. Common mistake is doing the dot product wrongly or keying into the calculator wrongly.

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(d)	Equation of plane $ABC : \mathbf{r} \cdot \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} = 15$ Equation of line $l : \mathbf{r} = \begin{pmatrix} 5 \\ -1 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ -2 \\ -13 \end{pmatrix}, \lambda \in \mathbb{R}$ At intersection, $\begin{pmatrix} 5-6\lambda \\ -1-2\lambda \\ 8-13\lambda \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ 2 \end{pmatrix} = 15$ $34 - 38\lambda = 15 \Rightarrow \lambda = \frac{1}{2}$ $\overrightarrow{OE} = \begin{pmatrix} 5 \\ -1 \\ 8 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -6 \\ -2 \\ -13 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ \frac{3}{2} \end{pmatrix}$ Hence coordinates of point $E \left(2, -2, \frac{3}{2} \right)$	This part is generally well-attempted.
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(e)



Let D' be the point of reflection of D in the mirror.

Using part(b),

$$\frac{\overrightarrow{OD'} + \overrightarrow{OD}}{2} = \frac{1}{22} \begin{pmatrix} 53 \\ 35 \\ 138 \end{pmatrix}$$

$$\overrightarrow{OD'} = \frac{1}{11} \begin{pmatrix} 53 \\ 35 \\ 138 \end{pmatrix} - \begin{pmatrix} 5 \\ -1 \\ 8 \end{pmatrix} = \frac{1}{11} \begin{pmatrix} -2 \\ 46 \\ 50 \end{pmatrix}$$

$$\overrightarrow{ED'} = \frac{1}{11} \begin{pmatrix} -2 \\ 46 \\ 50 \end{pmatrix} - \begin{pmatrix} 2 \\ -2 \\ 3/2 \end{pmatrix} = \begin{pmatrix} -24/11 \\ 68/11 \\ 67/22 \end{pmatrix} = \frac{1}{22} \begin{pmatrix} -48 \\ 136 \\ 67 \end{pmatrix}$$

Drawing a diagram to try to visualise the line of reflection would be the first step.

Students who make use of the result in the earlier part of the question will get the answer faster. Other methods of doing without using the earlier result will generally have longer working but get the correct answer, nonetheless.

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	Vector equation of line $\mathbf{r} = \begin{pmatrix} 2 \\ -2 \\ \frac{3}{2} \end{pmatrix} + \gamma \begin{pmatrix} -48 \\ 136 \\ 67 \end{pmatrix}, \gamma \in \mathbb{R}$	
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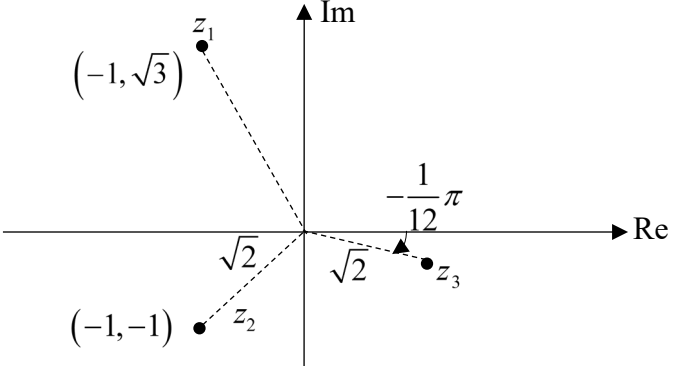
Q1	Suggested Answers	Guidance
(a)	$\alpha = -3, \beta = 1$	Mostly well done.
(b)	Area of loop $= 2 \int_{-3}^1 (1-x) \sqrt{x+3} \, dx$ $= 2 \int_0^2 (4-u^2) u \times 2u \, du$ $= 4 \int_0^2 (4u^2 - u^4) \, du$ $= 4 \left[\frac{4}{3} u^3 - \frac{1}{5} u^5 \right]_0^2 = \frac{256}{15}$ $u = \sqrt{x+3} \quad u = \sqrt{x+3}$ $u^2 = x+3 \quad \text{or} \quad \frac{du}{dx} = \frac{1}{2\sqrt{x+3}}$ $2u \frac{du}{dx} = 1 \quad \frac{du}{dx} = \frac{1}{2u}$ When $x = -3, u = 0$ When $x = 1, u = 2$	Generally, students can perform the substitution of the following 3 parts in the integral: $2 \int_{-3}^1 \underbrace{(1-x)}_3 \underbrace{\sqrt{x+3}}_1 \underbrace{dx}_2$ Reasons for not obtaining correct answers for some include: - not changing the limits (in part 3 above) to u-values, - using wrong answers from part (a), or - not reading question requiring exact area.

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Q2	Suggested Answers	Guidance
(a)	<u>Line segment</u> OA	Stating it is a line through O and A (or parallel to vector $\mathbf{a}$) is inadequate as the implication of $0 \leq \lambda \leq 1$ needs to be considered (i.e. line is finite in length).
(b)	<u>Line passing through</u> O parallel to AB	$\mathbf{r} \times (\mathbf{a} - \mathbf{b}) = \mathbf{0} \Rightarrow \overrightarrow{OR} \times \overrightarrow{BA} = \mathbf{0}$ Hence OR is parallel to AB (or BA), passing through O . It is insufficient to use vectors $\mathbf{r}$ or $(\mathbf{a} - \mathbf{b})$ in answers as the geometrical interpretation must be given. Some students misinterpreted and tried to link up part (a) of this question to conclude the result for $\mathbf{r}$ that must be satisfied.
(c)	It is a sphere (or circle) with OA as a diameter	Many students recognised the equation of a plane and merely stated results as “a plane passing through origin with normal $(\mathbf{r} - \mathbf{a})$ ” which is inappropriate as $\mathbf{r}$ represents the position vector of the varying point R . $\mathbf{r} \cdot (\mathbf{r} - \mathbf{a}) = 0 \Rightarrow \overrightarrow{OR} \cdot \overrightarrow{AR} = 0$ ie $\angle ORA = 90^\circ$ led many to conclude the result as a right-angled triangle ORA which does not correctly refer to the geometrical context that R is a variable point. Given O and A fixed, the possible positions of the varying point R is on a sphere (or circle) with OA as a diameter (geometry of angle in a semi-circle). Therefore, $\mathbf{r}$ is a set of position vectors on a sphere (or circle) with OA as a diameter. Again, some students misinterpreted and tried to link up parts (a) and (b) of this question to conclude the result for $\mathbf{r}$ that must be satisfied.
(d)	$(\mathbf{r} - \mathbf{a}) \times (\mathbf{r} - \mathbf{b}) = \mathbf{0}$ $(\mathbf{r} - \mathbf{a}) = \mu(\mathbf{r} - \mathbf{b})$ where $\mu \neq 1$ $\mathbf{r}(1 - \mu) = \mathbf{a} - \mu\mathbf{b}$	Many students ‘expanded’ the equation, ending up inability to simplify any further. Similar problem of some students misinterpreting and trying to link up all the previous parts of this question to conclude the result for $\mathbf{r}$ that must be satisfied when the question did not indicate so. Overall, question highlighted the weak vector concepts:

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$\mathbf{r} = \frac{\mathbf{a} - \mu \mathbf{b}}{(1 - \mu)} = \left(\frac{1}{1 - \mu} \right) \mathbf{a} + \left(\frac{-\mu}{1 - \mu} \right) \mathbf{b}, \text{ where } \mu \neq 1$ Alternatively, $(\mathbf{r} - \mathbf{a}) \times (\mathbf{r} - \mathbf{b}) = \mathbf{0}$ implies $\overrightarrow{AR}$ is parallel to $\overrightarrow{BR}$ (A, B and R are collinear) R lies on the line AB. $\mathbf{r} = \mathbf{a} + \alpha(\mathbf{b} - \mathbf{a})$, where $\alpha \in \mathbb{R}$	• $\mathbf{a} \cdot \mathbf{b} = 0$ indicates $\mathbf{a}$ and $\mathbf{b}$ perpendicular versus $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ indicating $\mathbf{a}$ and $\mathbf{b}$ parallel. • $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$ is a line containing points (R) with position vector $\mathbf{r}$ passing through a point (A) with position vector $\mathbf{a}$, parallel to vector $\mathbf{b}$.
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Q3	Suggested Answers	Guidance
(a)	$z_1 = 2e^{i\left(\frac{2}{3}\pi\right)}, z_2 = \sqrt{2}e^{i\left(-\frac{3}{4}\pi\right)}, z_3 = \sqrt{2}e^{i\left(-\frac{1}{12}\pi\right)}$ 	For z_1 & z_2, candidates can choose to either label the coordinates or the modulus & argument. Many candidates made the mistake of writing $z_3 = \sqrt{2}e^{i\left(\frac{1}{12}\pi\right)}$ Note that $z_3 = \sqrt{2}\left(\cos\frac{1}{12}\pi - i\sin\frac{1}{12}\pi\right)$ $= \sqrt{2}\left(\cos\frac{1}{12}\pi + i\sin\frac{1}{12}\pi\right)^*$ $= \sqrt{2}e^{i\left(-\frac{1}{12}\pi\right)}$
(b)	$z_1 = 2e^{i\left(\frac{2}{3}\pi\right)}, z_2 = \sqrt{2}e^{i\left(-\frac{3}{4}\pi\right)}, z_3 = \sqrt{2}e^{i\left(-\frac{1}{12}\pi\right)}$ Method 1: Properties of modulus & argument $\left \frac{z_1}{z_2(z_3)^2}\right = \frac{ z_1 }{ z_2 z_3 ^2} = \frac{2}{(\sqrt{2})(\sqrt{2})^2} = \frac{1}{\sqrt{2}}$ $\arg\left(\frac{z_1}{z_2(z_3)^2}\right) = \arg z_1 - \arg z_2 - 2\arg z_3$ $= \frac{2}{3}\pi - \left(-\frac{3}{4}\pi\right) - 2\left(-\frac{1}{12}\pi\right)$ $= \frac{19}{12}\pi$	Many candidates were unable to correctly obtain the argument of z_1 & z_2. Candidates should use the Argand diagram to guide them to see what the basic angle is and how to obtain the arguments correctly.

	$\therefore \arg \left(\frac{z_1}{z_2 (z_3)^2} \right) = \frac{19}{12} \pi - 2\pi = -\frac{5}{12} \pi$ $\therefore \frac{z_1}{z_2 (z_3)^2} = \frac{1}{\sqrt{2}} e^{i \left(-\frac{5}{12} \pi \right)}$	
(b)	<u>Method 2: Using exponential form</u> $\frac{z_1}{z_2 (z_3)^2} = \frac{2e^{i \left(\frac{2}{3} \pi \right)}}{\left(\sqrt{2} e^{i \left(-\frac{3}{4} \pi \right)} \right) \left(\sqrt{2} e^{i \left(-\frac{1}{12} \pi \right)} \right)^2}$ $= \frac{1}{\sqrt{2}} e^{i \left(\frac{19}{12} \pi \right)}$ $= \frac{1}{\sqrt{2}} e^{i \left(\frac{19}{12} \pi - 2\pi \right)}$ $= \frac{1}{\sqrt{2}} e^{i \left(-\frac{5}{12} \pi \right)}$	
(c)	Let $z_4 = r e^{i\theta}$ <u>Method 1: Properties of modulus & argument</u> $\left \frac{z_1 z_4}{z_2 (z_3)^2} \right = 1$ $\left \frac{z_1}{z_2 (z_3)^2} \right z_4 = 1$ $\Rightarrow r = \sqrt{2}$	

$\arg\left(\frac{z_1 z_4}{z_2 (z_3)^2}\right) = \arg\left(\frac{z_1}{z_2 (z_3)^2}\right) + \arg z_4$ $= -\frac{5}{12}\pi + \theta$ Since $\frac{z_1 z_4}{z_2 (z_3^*)^2}$ is purely real, $-\frac{5}{12}\pi + \theta = k\pi, k \in \mathbb{Z}$ $\theta = k\pi + \frac{5}{12}\pi$ $= \frac{5}{12}\pi \text{ or } -\frac{7}{12}\pi \quad (\because -\pi < \theta \leq \pi)$ $\therefore z_4 = \sqrt{2}\left(\cos\frac{5}{12}\pi + i\sin\frac{5}{12}\pi\right) \text{ or }$ $\sqrt{2}\left(\cos\left(-\frac{7}{12}\pi\right) + i\sin\left(-\frac{7}{12}\pi\right)\right)$	A large number of candidates incorrectly expressed the general form of $\arg\left(\frac{z_1 z_4}{z_2 (z_3)^2}\right)$
(c) <u>Method 2: Using exponential form</u> $\frac{z_1 z_4}{z_2 (z_3)^2} = \left[\frac{1}{\sqrt{2}} e^{i\left(-\frac{5}{12}\pi\right)}\right] (r e^{i\theta})$ $= \frac{r}{\sqrt{2}} e^{i\left(-\frac{5}{12}\pi + \theta\right)}$ $\frac{r}{\sqrt{2}} = 1 \Rightarrow r = \sqrt{2}$ Since $\frac{z_1 z_4}{z_2 (z_3^*)^2}$ is purely real, $-\frac{5}{12}\pi + \theta = k\pi, k \in \mathbb{Z}$ $\theta = k\pi + \frac{5}{12}\pi = \frac{5}{12}\pi \text{ or } -\frac{7}{12}\pi \quad (\because -\pi < \theta \leq \pi)$	

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	$\therefore z_4 = \sqrt{2} \left(\cos \frac{5}{12} \pi + i \sin \frac{5}{12} \pi \right) \text{ or}$ $\sqrt{2} \left(\cos \left(-\frac{7}{12} \pi \right) + i \sin \left(-\frac{7}{12} \pi \right) \right)$	
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Q4	Suggested Answers	Guidance
(a)	$\frac{1}{4r^2 - 8r + 3} = \frac{A}{2r - 3} + \frac{B}{2r - 1}$ $A(2r - 1) + B(2r - 3) = 1$ When $r = \frac{1}{2}$, $B = -\frac{1}{2}$ When $r = \frac{3}{2}$, $A = \frac{1}{2}$ $\frac{1}{4r^2 - 8r + 3} = \frac{1}{2} \left(\frac{1}{2r - 3} - \frac{1}{2r - 1} \right)$	Some form of working for the Partial Fractions should be shown as wrong values for the two unknowns warrant NO method marks otherwise.
(b)	$\sum_{r=2}^{3n} \left(\frac{1}{4r^2 - 8r + 3} \right) = \frac{1}{2} \sum_{r=2}^{3n} \left(\frac{1}{2r - 3} - \frac{1}{2r - 1} \right)$ $= \frac{1}{2} \left(\begin{array}{c} 1 - \frac{1}{3} \\ + \frac{1}{3} - \frac{1}{5} \\ + \frac{1}{5} - \frac{1}{7} \\ \vdots \\ + \frac{1}{6n-5} - \frac{1}{6n-3} \\ + \frac{1}{6n-3} - \frac{1}{6n-1} \end{array} \right)$ $= \frac{1}{2} \left(1 - \frac{1}{6n-1} \right) \quad \left(= \frac{3n-1}{6n-1} \right)$	Students need to show at least the first two and last two rows of cancellation clearly. Some substituted the values of r without evaluating and made mistakes along the way.

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(c)	$\sum_{r=2}^{\infty} \left(\frac{1}{4r^2 - 8r + 3} \right) = \frac{1}{2}$	Using this $\frac{1}{2} \left(1 - \frac{1}{6n-1} \right)$ is the most direct way to derive the answer. However, many wrote this as approaching the value instead of using the equal sign.
(d)	$\begin{aligned} \sum_{r=n+1}^{3n} \left(\frac{1}{4r^2 - 8r + 3} \right) &= \sum_{r=2}^{3n} \left(\frac{1}{4r^2 - 8r + 3} \right) - \sum_{r=2}^n \left(\frac{1}{4r^2 - 8r + 3} \right) \\ &= \frac{1}{2} \left(1 - \frac{1}{6n-1} \right) - \frac{1}{2} \left(1 - \frac{1}{2n-1} \right) \\ &= \frac{1}{2} \left(\frac{1}{2n-1} - \frac{1}{6n-1} \right) \\ &= \frac{4n}{2(2n-1)(6n-1)} \\ &= \frac{2n}{(2n-1)(6n-1)} \end{aligned}$	Many misread or ignore the “Using the result in part (b)” to attempt this part resulting in 0 mark. Method mark is only awarded based on clear working showing the use of the result in part (b). The final accepted answer should be a simplified fraction.

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Q5	Suggested Answers	Guidance
(a)	Volume of wok $= \pi \int_{-16}^q (256 - y^2) dy$ $= \pi \left[256y - \frac{y^3}{3} \right]_{-16}^q$ $= \pi \left[\left(256q - \frac{q^3}{3} \right) - \left(-4096 + \frac{4096}{3} \right) \right]$ $= \pi \left(256q - \frac{q^3}{3} + \frac{8192}{3} \right)$ $\pi \left(256q - \frac{q^3}{3} + \frac{8192}{3} \right) = 3300$ Using GC, $q = -7.01245634$ Depth of wok $-7.01245634 - (-16) = 9$ cm (correct to nearest integer)	Note: volume of wok is not volume of hemisphere. There are many WRONG formulations of volume of wok: (1) $-\pi \int_{-16}^q (256 - y^2) dy$ (2) $2\pi \int_{-16}^q (256 - y^2) dy$ (3) $\frac{\pi}{2} \int_{-16}^q (256 - y^2) dy$ (4) $\pi \int_{-16}^q (256 - x^2) dx$
(b)	Method 1 (direct integration): Volume of flat frying pan $= \pi \int_r^0 (256 - y^2) dy$ $= \pi \left[256y - \frac{y^3}{3} \right]_r^0$ $= \pi \left[0 - \left(256r - \frac{r^3}{3} \right) \right]$ $= \pi \left(\frac{r^3}{3} - 256r \right)$	

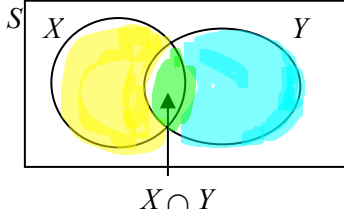
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	$\pi \left(\frac{r^3}{3} - 256r \right) = 1464\pi$ $\frac{r^3}{3} - 256r = 1464$ Using GC, $r = -6$	Convert the given capacity 1.464π litres to cm^3 .
(b)	Method 2 (using result from part (a)): Volume of flat frying pan $= \text{Volume of hemisphere} - \pi \left(256r - \frac{r^3}{3} + \frac{8192}{3} \right)$ $= \frac{2}{3}\pi(16)^3 - \pi \left(256r - \frac{r^3}{3} + \frac{8192}{3} \right)$ $= \pi \left(\frac{r^3}{3} - 256r \right)$ $\pi \left(\frac{r^3}{3} - 256r \right) = 1464\pi$ $\frac{r^3}{3} - 256r = 1464$ Using GC, $r = -6$	
(c)	Let time taken to fill the wok to full capacity be T seconds Method 1: $\frac{dV}{dt} = \frac{3}{55}t$ $\int dV = \int \frac{3}{55}t \, dt$	The rate $\frac{3}{55}t$ litres per second is not a constant. It is WRONG to write: $\text{Time taken} = \frac{3.3}{\frac{3}{55}t}$ The given rate is in litres per second. There is no need to convert 3.3 litres to cm^3

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$V = \frac{3t^2}{110} + C$ When $t = 0$, $V = 0$, $C = 0$ When $t = T$, $V = 3.3$, $3.3 = \frac{3T^2}{110}$ $T = 11$	
Method 2: $\frac{dV}{dt} = \frac{3}{55}t$ $\int_0^{3.3} dV = \int_0^T \frac{3}{55}t \, dt$ $[V]_0^{3.3} = \frac{3}{55} \left[\frac{t^2}{2} \right]_0^T$ $3.3 = \frac{3}{55} \left(\frac{T^2}{2} \right)$ $T = 11$	

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Q6	Suggested Answers	Guidance
(a)	Number of arrangements without restriction = $10! = 3628800$	The 2 red and 2 blue cars are distinct as they are of different makes. Treat the empty lot as one of the 10 distinct objects.
(b)	Number of arrangements with BB and RR together $= 8! \times 2! \times 2! = 161280$	
(c)	Method 1 Number of arrangements with B1B2 together = $9! \times 2! = 725760$ Similarly, number of arrangements with R1R2 together $9! \times 2! = 725760$ By complement, required number of arrangements is $= 3628800 - (725760 + 725760 - 161280)$ $= 2338560$	Using $n(X \cup Y)' = n(S) - n(X \cup Y)$ and $n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$ where event X is B1B2 together and event Y is R1R2 together. $X \cap Y$ is B1B2 and R1R2 are together.
(c)	Method 2 Number of arrangements with B1B2 together and R1R2 not together = $7! \times 2! \times {}^8C_2 \times 2! = 564480$ Similarly, number of arrangements with R1R2 together and B1B2 not together = 564480 By complement, required number of arrangements is $3628800 - 161280 - 564480 - 564480 = 2338560$	
(c)	Method 3 Number of arrangements with red cars separated (no restrictions on blue cars) = $8! \times {}^9C_2 \times 2! = 2903040$ Number of arrangements with red cars separated and blue cars together = $7! \times 2! \times {}^8C_2 \times 2! = 564480$ By complement, required number of arrangements is	9C_2 refers to the number of arrangements to insert the 2 red cars.

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	$2903040 - 564480 = 2338560$	
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Q7	Suggested Answers	Guidance																																																									
(a)	$P(S = s) > P(S = s + 1)$ $\frac{{}^{18}C_s {}^{12}C_{10-s}}{{}^{30}C_{10}} > \frac{{}^{18}C_{s+1} {}^{12}C_{9-s}}{{}^{30}C_{10}}$ $\frac{18!}{s!(18-s)!} \cdot \frac{12!}{(s+2)!(10-s)!} > \frac{18!}{(s+1)!(17-s)!} \cdot \frac{12!}{(s+3)!(9-s)!}$ $(s+1)!(17-s)!(9-s)!(s+3)! > s!(18-s)!(10-s)!(s+2)!$ $(s+1)(s+3) > (18-s)(10-s)$ $s^2 + 4s + 3 > s^2 - 28s + 180$ $32s > 177$ $s > 5.53$ Thus $s = 6$.	The approach of multiplying 10 probabilities together is not recommended. S is not a binomial random variable as the shapes are not chosen with replacement. Need to show the denominator for $\frac{{}^{18}C_s {}^{12}C_{10-s}}{{}^{30}C_{10}} > \frac{{}^{18}C_{s+1} {}^{12}C_{9-s}}{{}^{30}C_{10}}$ since $P(S = s)$ and $P(S = s + 1)$ are probabilities and this is a “show” question.																																																									
(b)	Outcome Table <table><tr><th colspan="2" rowspan="2">Absolute Difference</th><th colspan="6">No. on Square</th></tr><tr><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr><tr><td rowspan="4">No. on Triangle</td><td>1</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>2</td><td>1</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>3</td><td>2</td><td>1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td><td>1</td><td>2</td></tr></table> Probability Distribution <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$P(X = x)$</td><td>$\frac{4}{24}$</td><td>$\frac{7}{24}$</td><td>$\frac{6}{24}$</td><td>$\frac{4}{24}$</td><td>$\frac{2}{24}$</td><td>$\frac{1}{24}$</td></tr></table>	Absolute Difference		No. on Square						1	2	3	4	5	6	No. on Triangle	1	0	1	2	3	4	5	2	1	0	1	2	3	4	3	2	1	0	1	2	3	4	3	2	1	0	1	2	x	0	1	2	3	4	5	$P(X = x)$	$\frac{4}{24}$	$\frac{7}{24}$	$\frac{6}{24}$	$\frac{4}{24}$	$\frac{2}{24}$	$\frac{1}{24}$	An outcome table may be useful to identify the possible values of x correctly.
Absolute Difference				No. on Square																																																							
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No. on Triangle	1	0	1	2	3	4	5																																																				
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$P(X = x)$	$\frac{4}{24}$	$\frac{7}{24}$	$\frac{6}{24}$	$\frac{4}{24}$	$\frac{2}{24}$	$\frac{1}{24}$																																																					

(c)	$P(X_1 = 2 \mid X_1 + X_2 + X_3 = 3)$ $= \frac{P(X_1 = 2, X_2 = 0, X_3 = 1) \times 2!}{P(X_1 = 1, X_2 = 1, X_3 = 1) + P(X_1 = 1, X_2 = 2, X_3 = 0) \times 3! + P(X_1 = 3, X_2 = 0, X_3 = 0) \times \frac{3!}{2!}}$ $= \frac{\frac{6}{24} \left(\frac{4}{24} \right) \left(\frac{7}{24} \right) 2!}{\left(\frac{7}{24} \right)^3 + \frac{7}{24} \left(\frac{6}{24} \right) \left(\frac{4}{24} \right) 3! + \frac{4}{24} \left(\frac{4}{24} \right)^2 \frac{3!}{2!}}$ $= \frac{336}{1543} \text{ or } 0.218 \text{ (to 3 s.f.)}$	This is a conditional probability question.
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Q8	Suggested Answers	Guidance
(a)	Let L be the length of a randomly chosen rectangular cotton fabric. $L \sim N(24, 1.5^2)$ $P(L < 23.5) = 0.36944 \quad (5 \text{ s.f.})$ $= 0.369 \quad (3 \text{ s.f.})$	
(b)	Let B be the breadth of a randomly chosen rectangular cotton fabric. $B \sim N(20, 1.2^2)$ Perimeter $= 2L + 2B \sim N(88, 14.76)$ $P(L > 90) = 0.30133 \quad (5 \text{ s.f.})$ $= 0.301 \quad (3 \text{ s.f.})$	Since the opposite side are exact, the perimeter is $2L + 2B$, not $L_1 + L_2 + B_1 + B_2$
(c)	The length and breadth of each / a randomly chosen rectangular cotton fabric are independent of each other.	It is the length and the breadth of the same rectangular cotton fabric that is independent. Not across cotton fabrics.
(d)	$L \sim N(24, 1.5^2)$ $P(23 < L < 25) = 0.49502 \quad (5 \text{ s.f.})$ Let X be the number of rectangular cotton fabric with length between 23 and 25 cm, out of 48. $X \sim B(48, 0.49502)$ $E(X) = 48(0.49502) = 23.761 \quad (5 \text{ s.f.})$ $= 23.8 \quad (3 \text{ s.f.})$	Expected number of pieces here is the <u>average</u> number of pieces. <u>Average can have decimal places</u>. Students should not round up or round down the number 23.8.
(e)	$L_1 - L_2 \sim N(0, 4.5)$	Since the question said difference, then it is $L_1 - L_2$ and not $L_1 - L_2$.

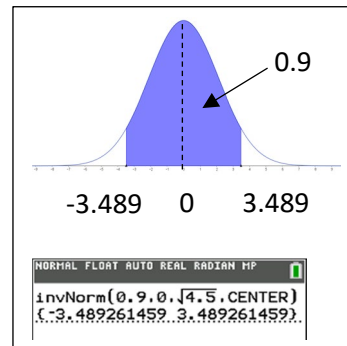
$$P(|L_1 - L_2| \leq k) \geq 0.9$$

$$P(-k \leq L_1 - L_2 \leq k) \geq 0.9$$

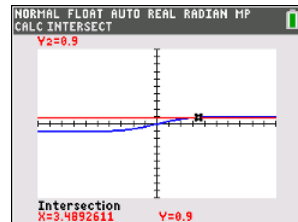
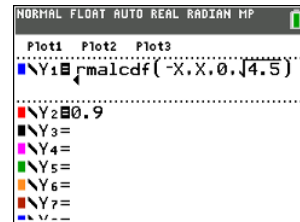
$$k \geq 3.4893$$

$$k \geq 3.49 \quad (3 \text{ s.f.})$$

Least $k = 3.49 \quad (3 \text{ s.f.})$



Method 2:



Method 3:

$$P(|L_1 - L_2| \leq k) \geq 0.9$$

$$P(-k \leq L_1 - L_2 \leq k) \geq 0.9$$

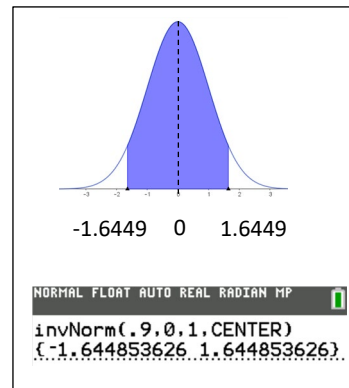
$$P\left(\frac{-k-0}{\sqrt{4.5}} \leq Z \leq \frac{k-0}{\sqrt{4.5}}\right) \geq 0.9$$

$$\frac{k}{\sqrt{4.5}} \geq 1.644853626$$

$$k > 3.48926$$

$$k \geq 3.49 \quad (3 \text{ s.f.})$$

Least $k = 3.49 \quad (3 \text{ s.f.})$



Also, student should draw a small bell-shaped curve, taking note of the mean and the probability of interest.

Common mistake include
take `invnorm(0.9,  $\mu$ ,  $\sigma$ , left)` to find k .

It should be

- (1) `invnorm(0.9,  $\mu$ ,  $\sigma$ , center)` or
- (2) `invnorm(0.95,  $\mu$ ,  $\sigma$ , left)`

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Q9	Suggested Answers	Guidance
(a)	Let X be the number of diners out of 10 who order the signature dish $X \sim B(10, 0.7)$ $P(X \geq 3) = 1 - P(X < 3)$ $= 1 - P(X \leq 2)$ $= 0.99841 \quad (5 \text{ s.f.})$ $= 0.998 \quad (3 \text{ s.f.})$	Key point is to rewrite $P(X \geq 3) = 1 - P(X \leq 2)$ to key into GC for the binomcdf command.
(b)	Method 1: $P(3 < X < 8) = P(X < 8) - P(X \leq 3)$ $= P(X \leq 7) - P(X \leq 3)$ $= 0.60663 \quad (5 \text{ s.f.})$ $= 0.607 \quad (3 \text{ s.f.})$	Many students made mistakes with the inequalities sign. “Between” does not include the 2 end-points.
(b)	Method 2 (not recommended): $P(3 < X < 8) = P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7)$ $= 0.60663 \quad (5 \text{ s.f.})$ $= 0.607 \quad (3 \text{ s.f.})$	
(c)	Let Y be the number of diners out of n who order the signature dish $Y \sim B(n, 0.7)$ $P(Y \leq 80) \geq 0.9$ largest $n = 106$ (maximum number of diners) 	We need to understand from context that the number of diners (trials) n must be more than 80 for $P(Y \leq 80)$ to be less than 1. If $n \leq 80$, then $P(Y \leq 80) = 1 \text{ (sure event).}$

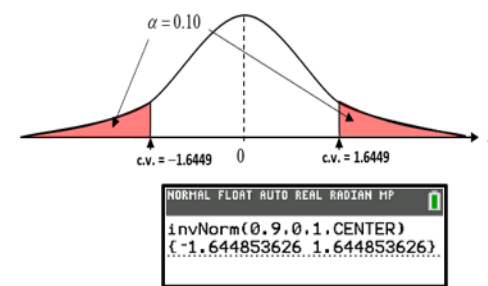
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(d)	$\left[P(X \geq 3) \right]^{40} = (0.99841)^{40}$ $= 0.938 \quad (3 \text{ s.f.})$ OR $T \sim B(40, 0.99841)$ $P(T = 40) = 0.93833 \approx 0.938$	Please use the 5 s.f. answer from part (a) to have the accuracy.
(e)	Let X be the number of diners out of 10 who order the signature dish, i.e. number of portions of the signature dish served at a randomly chosen table of 10 diners $X \sim B(10, 0.7)$ $E(X) = 7, \text{ Var}(X) = 2.1$ Average number of portions of the signature dish served per table, $\bar{X} = \frac{X_1 + X_2 + \dots + X_{40}}{40}$ Since $n = 40$ is large, by the Central Limit Theorem, $\bar{X} \sim N\left(7, \frac{2.1}{40}\right) \text{ approximately}$ $P(\bar{X} \geq 6.9) = 0.66874 \text{ (5 s.f.)}$ $= 0.669 \quad (3 \text{ s.f.})$	This part tests on the concept of taking Central limit theorem to get the distribution of $\bar{X} = \frac{X_1 + X_2 + \dots + X_{40}}{40}$ to be approximated to follow a normal distribution.

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Q10	Suggested Answers	Guidance
(a)	Unbiased estimate of population mean, $\bar{t} = \frac{543}{30} = 18.1$ Unbiased estimate of population variance, $s^2 = \frac{1}{29} \left[12722 - \frac{543^2}{30} \right] = 99.783 \quad (5 \text{ s.f.})$ $= 99.8 \quad (3 \text{ s.f.})$	
(b)	Let μ be the population mean bus arrival times after the scheduled pick-up time Test $H_0: \mu = 15$ $H_1: \mu > 15$ at 5 % level of significance Under H_0, since $n = 30$ is large, by the Central Limit Theorem, $\bar{T} \sim N\left(15, \frac{99.783}{30}\right)$ approximately $Z = \frac{\bar{T} - 15}{\sqrt{\frac{99.783}{30}}} \sim N(0, 1)$ approximately Test statistic: $z_{\text{cal}} = \frac{18.1 - 15}{\sqrt{\frac{99.783}{30}}} = 1.6998 \quad (5 \text{ s.f.}) \quad \text{p-value} = 0.044586$ Reject H_0 if $z_{\text{cal}} \geq 1.6449 \quad (5 \text{ s.f.}) \quad \text{p-value} \leq 0.05$ Since $z_{\text{cal}} \geq 1.6449 \quad (\text{p-value} \leq 0.05)$, there is sufficient evidence to reject H_0, i.e.	Note: question asks the student to work out the test statistic and use it to carry out the hypothesis test. The test statistic is $z_{\text{cal}} (=1.6998)$, and the student should compare z_{cal} with critical value. SR: No penalty for strict inequality sign when comparing test-statistic and critical value / p-value Award full credit if student uses $\bar{T} \sim N\left(15, \frac{99.783}{30}\right)$, and compare $\bar{t} = 18.1$ with c.v. = 18.0 to arrive at correct conclusion We can use the value of $\bar{t} (=18.1) > \mu (=15)$ to help confirm the direction of the inequality sign for the 1-tail test. Note that p-value must be small (<0.5). Students who obtained p-value = 0.095541 is not correct.

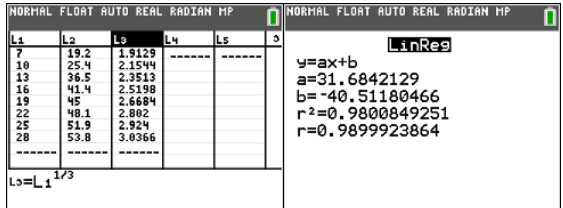
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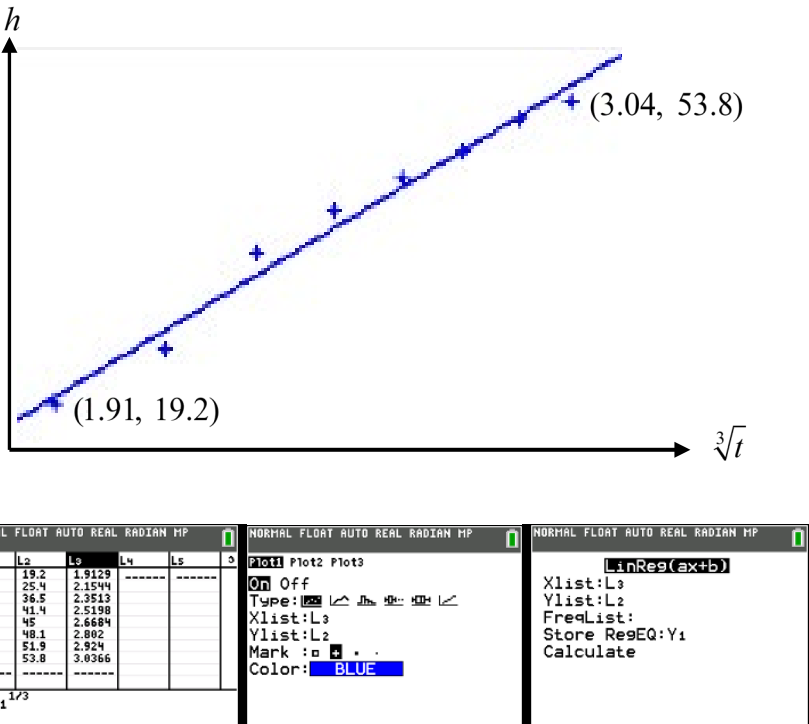
	there is sufficient evidence at 5% level of significance (to conclude) that the administration manager should agree with the feedback from teachers and students.	NORMAL FLOAT AUTO REAL RADIAN HP invNorm area:0.05 μ:15 σ:√(99.783/30) Tail: LEFT CENTER RIGHT Paste NORMAL FLOAT AUTO REAL RADIAN HP invNorm(0.05,15,√99.783/30 17.99981801						
(c)	Test $H_0 : \mu = 15$ $H_1 : \mu \neq 15$ at 10 % level of significance Under H_0, since $n = 40$ is large, by the Central Limit Theorem, $\bar{T} \sim N\left(15, \frac{99.783}{40}\right) \text{ approximately}$ $Z = \frac{\bar{T} - 15}{\sqrt{\frac{99.783}{40}}} \sim N(0, 1) \text{ approximately}$ Test statistic: $z_{\text{cal}} = \frac{\bar{t} - 15}{\sqrt{\frac{99.783}{40}}}$ Reject H_0 if $z_{\text{cal}} \leq -1.6449$ or $z_{\text{cal}} \geq 1.6449$ (5 s.f.) Since H_0 is not rejected $-1.6449 < z_{\text{cal}} < 1.6449$ $-1.6449 < \frac{\bar{t} - 15}{\sqrt{\frac{99.783}{40}}} < 1.6449$ $12.402 < \bar{t} < 17.598 \quad (5 \text{ s.f.})$ $12.4 < \bar{t} < 17.6 \quad (3 \text{ s.f.})$	 Note: Use non-strict inequality for rejecting H_0 and strict inequality for not rejecting H_0: <table><tr><th>Reject H_0</th><th>Do not reject H_0</th></tr><tr><td>$z_{\text{cal}} \leq -1.6449$ or $z_{\text{cal}} \geq 1.6449$</td><td>$-1.6449 < z_{\text{cal}} < 1.6449$</td></tr><tr><td>$p\text{-value} \leq 0.05$</td><td>$p\text{-value} > 0.05$</td></tr></table> Here, we do not use p-value since $\bar{t}$ is unknown.	Reject H_0	Do not reject H_0	$z_{\text{cal}} \leq -1.6449$ or $z_{\text{cal}} \geq 1.6449$	$-1.6449 < z_{\text{cal}} < 1.6449$	$p\text{-value} \leq 0.05$	$p\text{-value} > 0.05$
Reject H_0	Do not reject H_0							
$z_{\text{cal}} \leq -1.6449$ or $z_{\text{cal}} \geq 1.6449$	$-1.6449 < z_{\text{cal}} < 1.6449$							
$p\text{-value} \leq 0.05$	$p\text{-value} > 0.05$							
(d)	The sample of 40 buses is a random sample .							

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	There is no change in the unbiased estimate of the population variance bus arrival times after the scheduled pick-up time after the bus company claims to have made changes to its operation.	Many students confused the conditions for a random sample, for a binomial distribution, for calculating variance. One assumption is that the <u>sample was randomly chosen</u>. Students simply need to state that the sample of 40 buses was a random sample. There is no need to state the properties of a random sample (see first column of table below). Many students also confused the conditions for a random sample, for a binomial distribution, for calculating variance. <table> <tr> <th data-bbox="792 576 1205 616">Random sample</th><th data-bbox="1205 576 1617 616">Binomial distribution`</th><th data-bbox="1617 576 2033 616">Calculating variance</th></tr> <tr> <td data-bbox="792 616 1205 948"> <i>Every</i> bus arrival time has equal chance of being <i>selected</i>. The <i>selection</i> of a bus arrival time to be included in the sample is independent of the <i>selection</i> of other bus arrival times. </td><td data-bbox="1205 616 1617 948"> The probability a bus is late is constant for all buses. (Note: This need not be true.) The event that a bus is late is independent of the event other buses are late. (Note: This need not be true.) </td><td data-bbox="1617 616 2033 948"> The bus arrival times after the scheduled arrival time are independent of one another and are identically distributed. (The T_i's are i.i.d.) <u>Note that the above is implied when a sample is random.</u> </td></tr> </table>	Random sample	Binomial distribution`	Calculating variance	<i>Every</i> bus arrival time has equal chance of being <i>selected</i>. The <i>selection</i> of a bus arrival time to be included in the sample is independent of the <i>selection</i> of other bus arrival times.	The probability a bus is late is constant for all buses. (Note: This need not be true.) The event that a bus is late is independent of the event other buses are late. (Note: This need not be true.)	The bus arrival times after the scheduled arrival time are independent of one another and are identically distributed. (The T_i's are i.i.d.) <u>Note that the above is implied when a sample is random.</u>
Random sample	Binomial distribution`	Calculating variance						
<i>Every</i> bus arrival time has equal chance of being <i>selected</i>. The <i>selection</i> of a bus arrival time to be included in the sample is independent of the <i>selection</i> of other bus arrival times.	The probability a bus is late is constant for all buses. (Note: This need not be true.) The event that a bus is late is independent of the event other buses are late. (Note: This need not be true.)	The bus arrival times after the scheduled arrival time are independent of one another and are identically distributed. (The T_i's are i.i.d.) <u>Note that the above is implied when a sample is random.</u>						

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Q11	Suggested Answers	Guidance
(a)	Since $(\bar{t}, \bar{h})$ lies on the regression line, $\bar{h} = 1.6393\bar{t} + 11.475$ $\frac{k + 273.2}{8} = 1.6393(17.5) + 11.475$ $k = 48.1 \text{ (to 1 d.p.)}$	We cannot substitute $t=22$ into the equation as the data point may not lie on the regression line. The only point that will pass through the regression line is $(\bar{t}, \bar{h})$. As this is a show question, there must be clear working shown to get the answer,
(b)	For $h = a\sqrt[3]{t} + b$, $r = 0.98999$ For $h = c\sqrt{t} + d$, $r = 0.98615$ Since r for $h = a\sqrt[3]{t} + b$ is closer to 1, the linear correlation between h and $\sqrt[3]{t}$ is stronger. Using GC, the equation is $h = 31.684\sqrt[3]{t} - 40.512$ (5 s.f.) $h = 31.7\sqrt[3]{t} - 40.5$ (3 s.f.) 	Always check that the values keyed into GC is correct before proceeding. Leave all non-exact answers to 5 s.f. first before giving a 3 s.f. answer (unless the question requires otherwise). There is no need to write down too many decimal places.
(c)	Since the age of 2 months ($t \approx 60$) is out of the data range of t, the estimate is not reliable.	Be mindful of the units.

(d)		As is a h against $\sqrt[3]{t}$ scatter plot, the horizontal axis values must be labelled correctly. E.g. when $t = 7$, $\sqrt[3]{7} = 1.91$ (3s.f.) There is a need to specifically label the end values so that we are aware of the data range. Regression line with correct relative position with points. Points 1, 2, 8 below Points 3, 4, 5 above Points 6 and 7 to lie very close to the line (more accurately below the line)
(e)	Value of residual = $\underbrace{53.8}_{\text{Actual value of } h} - \underbrace{(31.684\sqrt[3]{28} - 40.512)}_{\text{Predicted value of } h \text{ using the regression line}} = -1.90 \text{ (3 s.f.)}$	In the question, it was mentioned that the value of the residual (can be positive or negative) is $q-f(p)$. Many students gave the absolute value of the answer.
(f)	The values of the <u>residuals could be positive or negative</u>, and <u>adding them up might cause the values to cancel out</u>. <u>By squaring the residuals, all the values will be positive</u>.	Many wrong concepts seen either through poor permutation of words or wrong idea: “Square of sum of residuals” is wrong. This means you sum the residuals first then square which is different mathematically.

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	The <u>sum of squares of residuals</u> has to be <u>minimised</u> in order to find the least squares regression line.	“Least <u>number</u> of squares” is wrong. We don’t count the squares on the graph paper. “Equal number of positive and negative residuals” may not hold true for some best fit lines, like the one in this question.
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